

Analog IC Design Implementation and Verification

Week 10 – Introduction to Layout Design Techniques

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Overview of Layouts

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Module Objectives

In this module, you learn how to

- Identify the various physical implementation techniques, manual and automated, that are used for analog, mixed-signal, and digital design.
- Identify the differences between an analog-on-top and digital-on-top mixed-signal flow.



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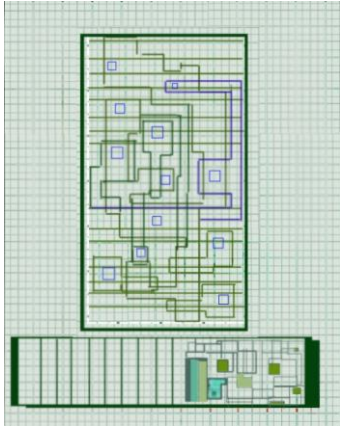


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History: Layout Design

In early days, integrated circuit (IC) layouts were drawn with pencils on mylar.

- The layouts were drawn by putting together a set of polygons.
- Originals were made by transferring the drawings from mylar to rubylith.
- Masks were made by photographic reduction from the rubylith original.



Pencil used for drawing layout on mylar

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In early days the layouts were hand drawn with a pencil over a mylar.

The first masks were made by transferring the drawings on the mylar to Rubylith.

History: What Is Rubylith?

Rubylith is a brand of masking film, invented and trademarked by Ulano.

- It consists of two films sandwiched together.
 - The bottom layer is a clear polyester backing sheet.
 - The top layer is a translucent, red, self-adhesive emulsion.
- The first microprocessor, the Intel 4004, was designed by physically cutting sheets of rubylith into thin strips to lay out the devices.

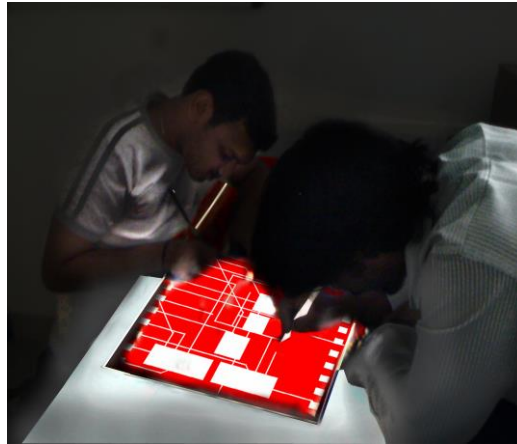


Picture : TTM/THIT 2008

Rubylith is a brand of masking film, invented and trademarked by the Ulano Corporation. Today the brand has become genericized to the point that it has become synonymous with all colored masking films. Rubylith consists of two films sandwiched together. The bottom layer is a clear polyester backing sheet. The top layer is a translucent, red colored, self-adhesive, emulsion. It is designed to be both easy to cut with a knife and light safe for use with orthochromatic films. Rubylith is used in many areas of graphic design, typically to produce masks for various printing techniques. For example, it is often used to mask off areas of a design when using a photoresist to produce printing plates for offset lithography or gravure. It is also frequently used during screen-printing. Ulano also produces a yellow colored, masking film called Amberlith®, that is not light safe but easier to use for masking when not employing a photoresist. The first microprocessor, the Intel 4004, was designed by physically cutting sheets of rubylith into thin strips to lay out the circuits to be printed, a process made virtually obsolete by current computer's graphic design capabilities.

Using Rubylith in Layout Design

After checking for design-rule violations, the silicon die was fabricated using masks that had been photographically reduced from the peeled rubylith originals.



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Rubylith is a two-layered material, which comes in huge sheets. The base layer is heavy transparent dimensionally stable mylar. A thin film of deep red cellophane-like material covers the base layer. The first chips at Intel used a machine called a "Coordinatograph" to guide cutting of the ruby layer. The coordinates and lengths had to be measured and transferred by hand to the cutter. Later, a Xynetics plotter with knives, instead of pens, was used to cut more quickly and precisely.

When the cutting was finished, the technicians had to peel away only the desired geometries that made the mask layers. The design engineer and mask designers would spend days hand-checking the rubylith for peeling errors, nicks, and unintended cuts. A final check was made for design rule violations. The rubylith was sent to the mask vendor to be made into masks for fabricating the silicon die.

Missing a cut or forgetting to peel a geometry would mean a bad part. The first Intel product, the 3101 was 64-bit RAM. Actually, the first version was only a 63-bit RAM due to a simple error peeling one layer on the rubylith.

The rubylith sheets had to be handled very carefully so they were not damaged. Small areas of ruby could be rubbed off. If there was any unexplained piece of ruby stuck in a random place on the mylar. This would solved after a several hour check against the layout to find if there were any missing bits and the mask was taken.

Adding or removing transistors and interconnect on rubylith was definitely a manual task, not unlike surgery. In fact, the technician who did the edits used a surgical scalpel and a metal ruler (scale). Adding transistors or interconnect involved cutting and peeling away bits of ruby. Removing objects involved adding ruby-red tape to the back of the heavy mylar. Cuts had to be precise so as to leave no nicks or cut marks on the mylar that might show on the mask. Verification was done with the metal scale and a 7X-magnifying eyepiece with a calibrated scale on the bottom."

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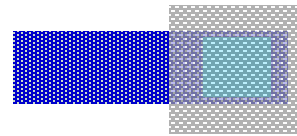


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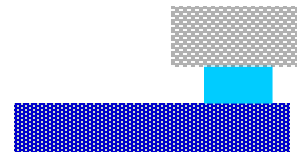
Polygon Pushing

The main mission of layout design is to define an efficient and effective physically integrated structure of electronic circuits.

- The physical structure can be defined in terms of geometric figures in different physical layers.
- The geometric figures are defined in terms of polygons that can be handled efficiently using computers.
- The simplest form of layout design is building every structure from scratch, by manually defining its polygons. This process is called *polygon pushing*, even when using layout software.



Top view



Cross-sectional view

Picture : TTM/THIT 2008

Physical structure can be defined in terms of geometric figures in different physical layers.

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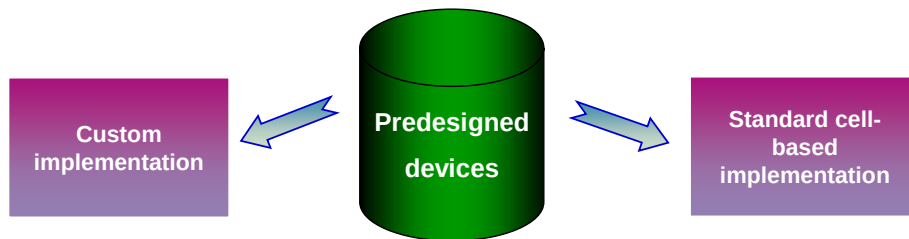
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What Are Predesigned Devices?

- Predesigned devices can be fixed-geometry devices that can be used in a design.
- Frequently used layouts can be drawn once and made reusable.
- Using these predesigned devices saves both time and disk space.
- Examples: Fixed-size bipolar transistors, diodes and CMOS, such as electrostatic discharge (ESD) devices.



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Picture: TTM/THIT 2008

Predesigned devices are manually generated layouts or devices which are further used as cell libraries or programmable devices in designing an ASIC (basically known as semi-custom ASIC).

For example, cell libraries in a standard cell-based ASIC design. The design differentiation and customization is achieved by creating custom metal layers that create custom connections between predefined lower-layer logic elements. These pre-defined lower-layer elements are also known as predesigned IP cores.

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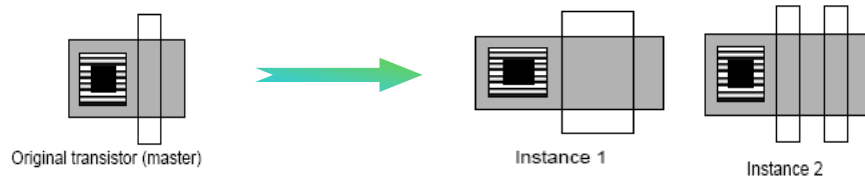


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Introduction to Parameterized Devices

A parameterized cell (pcell) is a programmable cell that lets you create a customized instance. It contains a layout view, a symbol view, a schematic view, and a behavioral view.

- A pcell master is created with SKILL® programming code which captures its structure and parameters.
- Different instances of a single pcell can represent forms and sizes of the design component.
- There are different ways of creating pcells.
 - Graphically generated pcells
 - SKILL-based pcells



Pictures : Cadence PCellref.pdf

Pcell is an abbreviation for parametrized cell, a concept used in the design of analog (or custom digital) layout. A pcell is an automatically generated graphical model (cell) representing a physical structure inside an integrated circuit (IC).

Structures within an IC and the rules (design rules) governing their physical dimensions are often complex making the structures tedious to draw by hand. Using a pcell, a circuit designer need only enter a few parameters and an electronic design automation (EDA) software tool draws the structure automatically.

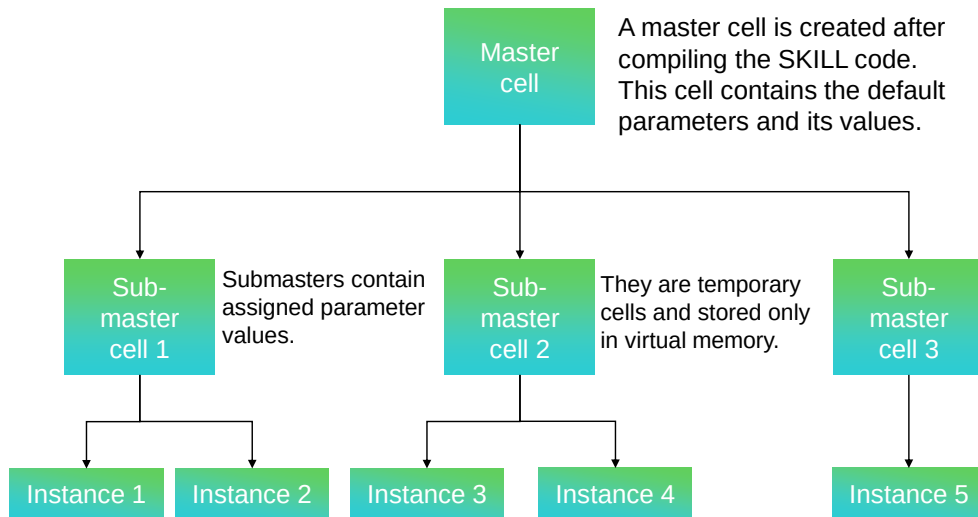
Advantages of Pcells

- Pcells are correct by design and reduce the verification effort.
Pcell use eliminates errors that can result from maintaining multiple versions of the same cell.
- Pcells speed up the process of entering layout data by eliminating the need to create duplicate versions of the same functional part.
- Pcells eliminate the need to explode levels of hierarchy when you want to change a small detail of a design.
- Pcells save disk space by creating a library of cells for similar parts that are all linked to the same source.



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Pcell Masters and Submasters



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Picture: TTM/TIIT 2008 (reference cadence PCellref.pdf)

Pcell is just created as any other layout and the parameters are defined. The Pcell which is created in that way is called a master. A master is the combination of the graphic layout you draw to make a cell and the parameters you assign to it. After you compile the master, it is stored in the database in the form of a SKILL procedure. The edits you make to the master appear in the cell instances when you compile the master. You must make all changes to the master. You cannot edit an instance of a Pcell.

By the way, sub masters are temporary cells created by the system in virtual memory and they are never stored in the disk.



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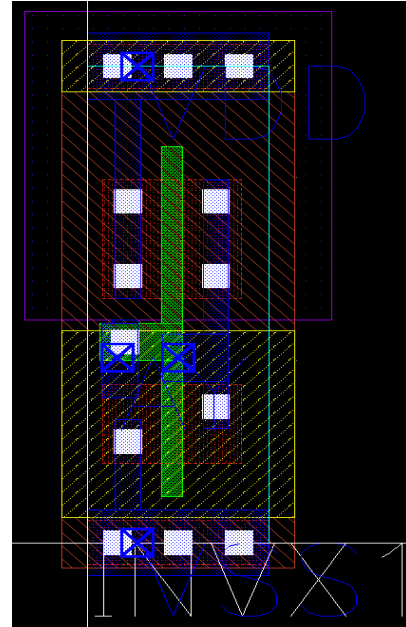


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What Are Standard Cells?

Standard cells are devices grouped and connected to provide a specific standard function, usually digital.

- Usually standard cells have variable width, but constant height, to allow them to be easily lined up in rows.
- Typical examples include cells with Boolean logic functions such as AND, OR, XOR, XNOR, inverters, and storage functions such as flip-flop and latch.



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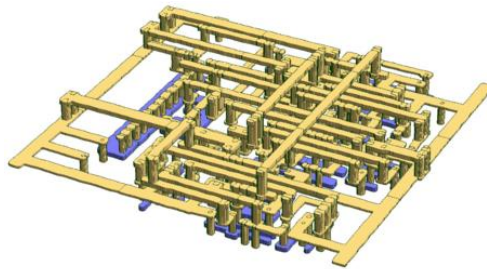
Picture: Snapshot of INVX1 from Cadence gpd90

In semiconductor design, *standard cell methodology* is a method of designing Application Specific Integrated Circuits (ASICs) with mostly digital-logic features. Standard cell methodology is an example of design abstraction, whereby a low-level VLSI-layout is encapsulated into an abstract logic representation (such as a NAND gate). Cell-based methodology (the general class that standard-cell belongs to) makes it possible for one designer to focus on the high-level (logical function) aspect of digital-design, while another designer focused on the implementation (physical) aspect. Along with semiconductor manufacturing advances, standard cell methodology was responsible for allowing designers to scale ASICs from comparatively simple single-function ICs (of several thousand gates), to complex multi-million gate devices (SoC).

Construction of a Standard Cell

Design of a standard cell is performed at the transistor level.

- CAD software such as Spectre® Circuit Simulator is used to simulate the electronic behavior and the performance of the circuit.
- Like pcells, standard cells include several views. These views include schematic, netlist, symbol, noise model, Verilog®, and a timing view (.lib). The last two are used by digital simulators.
- Standard cells are mostly used in application specific integrated circuit (ASIC) design flows (synthesis, place and route).



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A standard cell is group of transistor and interconnect structures, which provides a Boolean logic function (e.g., AND, OR, XOR, XNOR, inverters) or a storage function (flipflop or latch). The simplest cells are direct representations of the elemental NAND, NOR, and XOR boolean function, although cells of much greater complexity are commonly used (such as a 2-bit full-adder, or muxed D-input flipflop.) The cell's boolean logic function is called its *logical view*: functional behavior is captured in the form of a truth table or Boolean algebra equation (for combinational logic), or a state transition table (for sequential logic).

Usually, the initial design of a standard cell is developed at the transistor level, in the form a transistor netlist. The netlist is a nodal description of transistors, of their connections to each other, and their terminals (ports) to the external environment. Designers use Computer Aided Design (CAD) programs such as SPICE to simulate the electronic behavior of the netlist, by declaring input stimulus (voltage or current waveforms) and then calculating the circuit's time domain (analogue) response. The simulations verify whether the netlist implements the requested function, and predict other pertinent parameters such as power consumption or signal propagation delay.

Since the logical and netlist views are only useful to abstract (algebraic) simulation, and not device fabrication, the physical representation of the standard cell must be designed too. Also called the *layout view*, this is the lowest level of design abstraction in common design practice. From a manufacturing perspective, the standard cell's VLSI layout is the most important view, as it is closest to an actual "manufacturing blueprint" of the standard cell. The layout is organized into *base layers*, which correspond to the different structures of the transistor devices, and *interconnect lines*, which join together the terminals of the transistor formations.

For a typical Boolean function, many different transistor netlists exist that are functionally equivalent. Likewise, for a typical netlist, there exist many different layouts that fit the netlist's performance parameters. The designer's challenge is to minimize the manufacturing cost of the standard-cell's layout (generally by minimizing the circuit's die area), while still meeting the cell's speed and power performance requirements. Consequently, integrated circuit layout is a highly labor intensive job, despite the existence of design tools to aid this process.

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"<http://en.wikipedia.org/wiki/Image:Eda-fabrication.PNG>"

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Physical Implementation Techniques

- Custom implementation

For example, amplifiers, PLLs, mixers, ...

- Custom analog/mixed-signal/RF

Implementation is based on schematics, and optimized manually to meet layout constraints, and specifications in SPICE-based and mixed-signal testbenches.

- Custom digital

Implementation is based on the synthesized netlist, and layout is optimized manually to meet timing constraints, and functional specifications contained in logic testbenches.

- ASIC digital implementation:

For example, graphical functions, processors, ...

Implementation is based on the synthesized netlist, and placement and routing is machine-optimized to meet timing constraints, and functional specifications contained in logic testbenches.

- Other implementation techniques

For example, communication chips, ...

- Memory compilation
 - Datapath generation
 - Structured ASIC

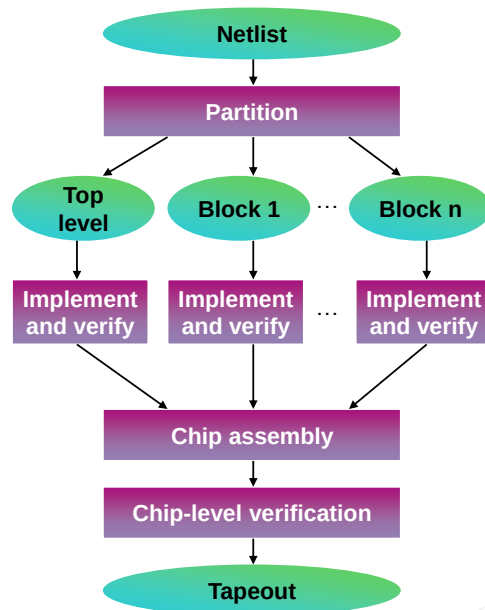


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Flat vs. Hierarchical Implementation

Chips can be laid out *flat* or *hierarchically*.

- In hierarchical implementation flows (see diagram), a design is partitioned into a top level and several blocks.
- Each block is implemented and verified separately.
- The top-level design uses abstracts of the blocks, until all blocks are ready.
- The process of integrating the blocks into the top-level design is called *chip assembly* or *integration*.
- Final verification precedes tapeout.



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Custom Implementation Methodologies

There are two basic methodologies of custom layout design.

- Manual (Virtuoso® Layout Editor): polygon pushing
 - In this method every design component is designed from scratch by pushing polygons with basic polygon commands.
 - Drawn polygons do not carry any connectivity information.
 - Predefined devices do not contain connectivity information. They are just a collection of polygons that can be dropped into a layout.
- Automated (Virtuoso XL Layout Editor): connectivity-driven design
 - In automated layouts, every polygon that is placed in the design canvas contains the connectivity information of the source schematics.
 - These polygons aid automated placers or routers to identify the source and destinations Virtuoso Custom Placer and Virtuoso Chip Assembly Router.
 - Predefined devices contain connectivity information, some even with parameterization capability.



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Connectivity-Driven Implementation

Connectivity-driven implementation is software-assisted generation of layouts, the connectivity of which is derived from a source. It is in contrast to manual layout that is designed by hand.

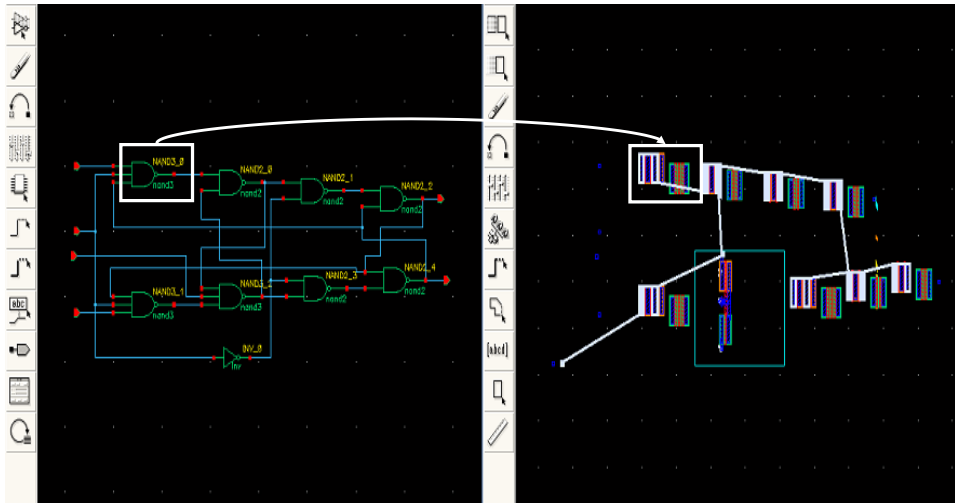
- The source is a schematic or a netlist, such as component description language (CDL) or SPICE simulator program.
- A layout-versus-schematic-correct (LVS-correct) layout can be designed in real time (using Virtuoso XL design tools).
- Advantages
 - Correct by design
 - Improved productivity
 - Reduced verification time



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Example: Connectivity-Driven Implementation

This shows the correspondence of the schematic components to the layout components and their connectivity.



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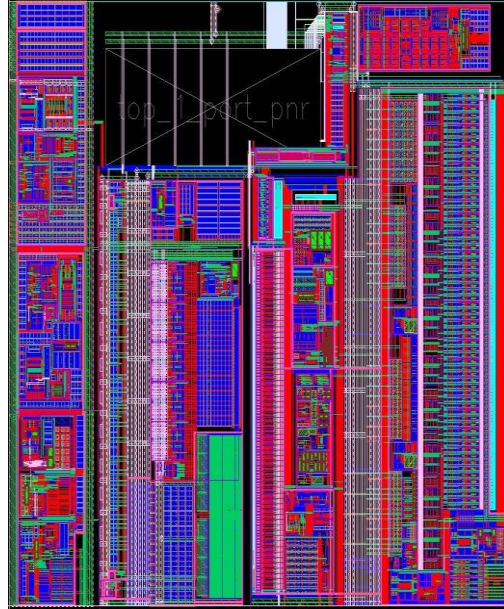
Picture : Snapshot from Cadence Virtuoso® XL Layout Editor Lab database

Virtuoso XL Layout Editor has set the standard and changed the way custom block authoring is done. Driven by a schematic connectivity source using Virtuoso Schematic Editor or a netlist source such as CDL or SPICE, an LVS-correct layout can be done in real-time to promote correct-by-construction layout, improved productivity, and reduced verification time. Tedious design tasks, such as device generation, placement, and routing, can be automated. Schematic and layout can be cross probed to highlight instances and devices and quickly identify unconnected nets.

Analog-On-Top Flows

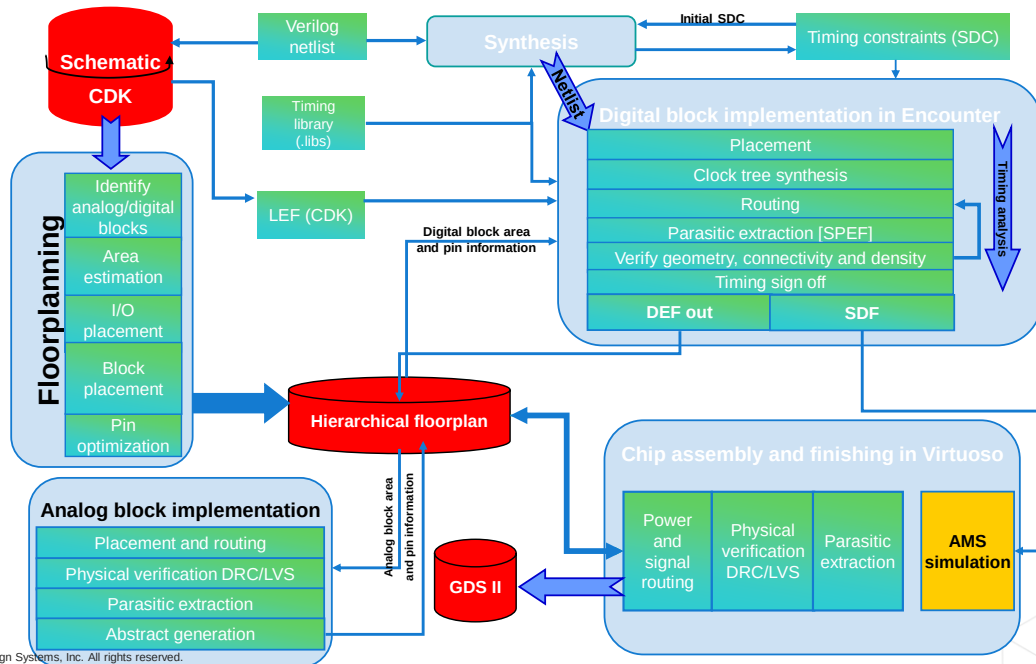
In analog-on-top (AOT) designs, the chip is mostly analog, but has few digital blocks, for which the top-level floorplan is implemented in a custom IC design environment.

- Many mixed-signal designs follow this technique for their full-chip physical implementation, for example, Ethernet PHY chips and so forth.
- AOT designs use the Virtuoso design platform as the cockpit and Encounter® IC design tools to design the digital blocks. For example, Virtuoso Digital Implementation Option is used for small digital blocks.



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Analog-On-Top Design Flow



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Standard Cell-Based Implementation

Standard cell design is a methodology for designing digital ICs using highly automated flows for placement and routing. It is known also as [application-specific integrated circuit](#) (ASIC) design.

- ASIC design usually requires much less manual optimization than a custom design.
- It shortens the design cycle by using standard cells, from which large logic circuits are constructed.
- While synthesis takes care of functionally correct netlist generation, the timing is often highly impacted by the placement and routing of cells. ASIC design involves automated algorithms to help meet timing constraints. Therefore, ASIC design is often called [timing-driven implementation](#).
- Generally, cells vary in width but are the same height. This helps the place-and-route tools in efficient placement (rows of equal-height standard cells, separated by routing channels). Tools such as the Encounter IC design platform also offer multi-height cell placement.



A standard cell design approach makes it possible to globally apply advanced optimization algorithms, which reduce the manual effort required and improve the quality of the synthesized logic during layout. The use of basic standard cell elements reduces complexity to the extent that a complete chip design can be handled flat by layout and test generation tools. This approach uses a small number of custom logic macros and custom memory arrays whenever a standard cell solution is not competitive. The major part of the combinational logic portions, however, are implemented in standard cells.

For example, cell libraries in a standard cell-based ASIC design. The design differentiation and customization is achieved by creating custom metal layers that create custom connections between predefined lower-layer logic elements. These pre-defined lower-layer elements are also known as predesigned IP cores.

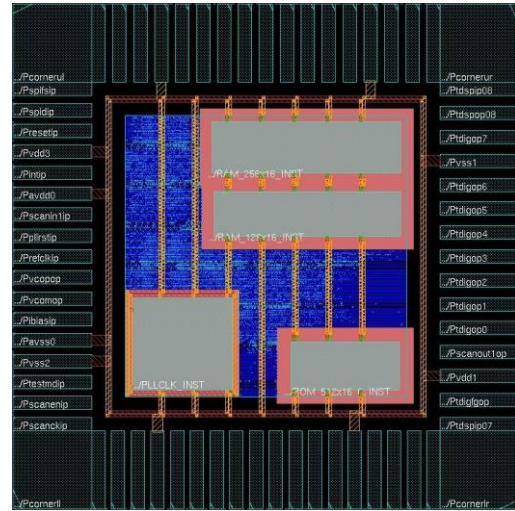
Timing-Driven Implementation

- Timing-driven implementation can be highly automated, even for complex designs, for example, system-on-a-chip (SoC).
- The timing constraints are communicated to the implementing tool through a standard timing constraint file, for example, Synopsys Design Constraints (SDC).
- In this approach, cells or nets with the most critical timing constraints are identified and placed or routed with priority.
- A lower priority is given to the noncritical cells and nets, which allows reducing the overall congestion of the design.



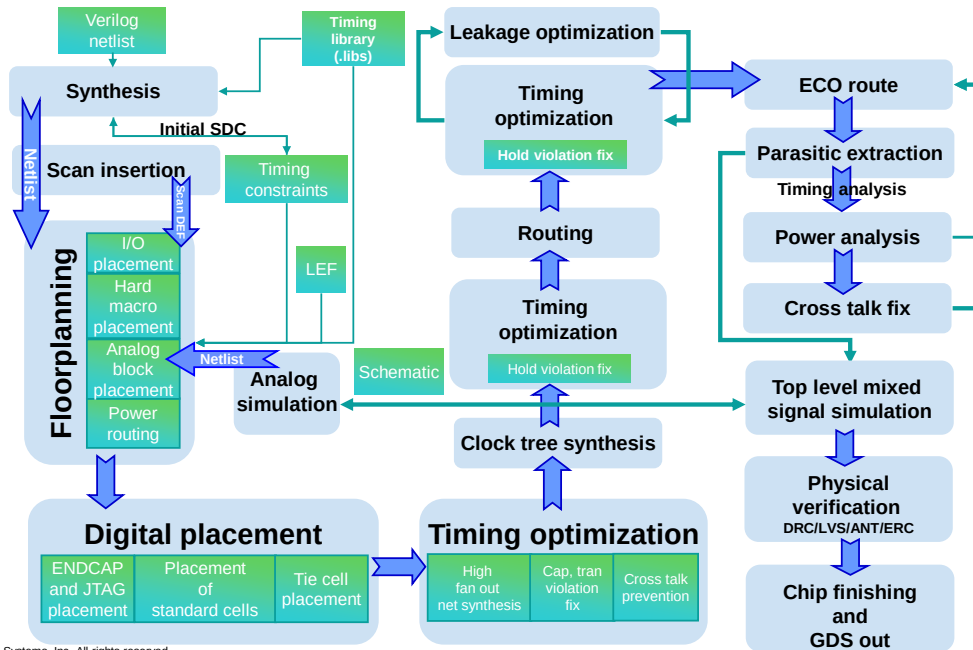
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DOT designs use the Encounter IC design platform as the cockpit and Virtuoso layout tools to design the analog blocks.



Picture: Snapshot of DTMP Chip design from Encounter 6.2 labs

Digital-On-Top Design Flow



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Layouts and Connectivity

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Module Objectives

In this module, you learn how to

- Identify the layers in a typical mixed-signal CMOS process, and their uses.
- Explain why design rules are needed and identify the purpose of some of the most fundamental design rules.



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Critical Layers of an IC

Integrated circuit layers

- Substrate
- Well (N-/P-)
- Poly
- Diffusion N+/P+
- Deep N-well
- Trench isolation
- Contact
- Metals
- Vias



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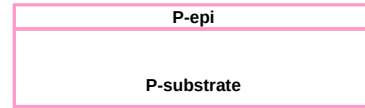
Critical Layers: Substrate

Substrate or body

- Substrate is a thin slice of semiconductor material upon which semiconductor devices are fabricated using photolithography.
- CMOS circuits are fabricated on a lightly doped p-type substrate.
- A thin epitaxial layer is deposited over the substrate to attain uniformity over the crystalline substrate.



Top view



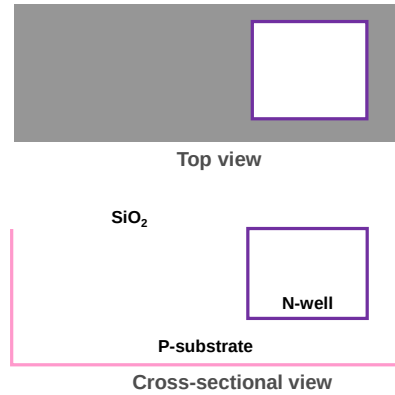
Cross-sectional view



Epitaxy or **epitaxial growth** is the process of depositing a thin layer (0.5 to 20 microns) of single crystal material over a single crystal substrate, usually through [chemical vapor deposition](#) (CVD). In semiconductors, the deposited film is often the same material as the substrate, and the process is known as homoepitaxy, or simply, epi. An example of this is silicon deposition over a silicon substrate.

Wells (N-/P-)

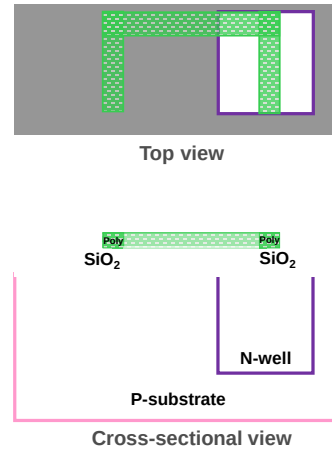
- Act as intermediate substrates for the respective MOS devices of semiconductor material. These wells are formed with diffusion or ion implantation.
- Are lightly doped with n-type or p-type impurities.



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Polysilicon

- Is widely used as MOS transistor gate electrodes and for interconnection in MOS circuits. It is also used as a resistor, as well as in ensuring ohmic contacts for shallow junctions.
- Can have a metal (such as tungsten) or metal silicide (such as tantalum silicide) deposited over it to enhance its conductivity when used as gate electrodes.



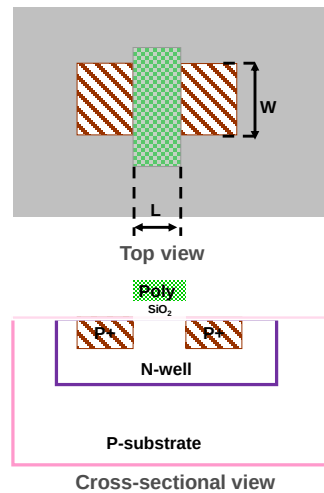
Gate A transistor that is formed when poly dissects diffusion. A gate is a device

Poly-Si is known to be compatible with high temperature processing and interfaces very well with thermal SiO₂. As a gate electrode, it has also been proven to be more reliable than Al. It can also be deposited conformally over steep topography. Heavily-doped poly thin films can also be used in emitter structures in bipolar circuits. Lightly-doped poly films can also be used as resistors.

Poly-Si is usually deposited by thermal decomposition or pyrolysis of silane at temperatures from 580-650 degrees C, with the deposition rate exponentially increasing with temperature. The deposition rate is also affected by the pressure of silane, which translates to silane concentration. Other important variables in [polysilicon deposition](#) are pressure and dopant concentration.

Diffusion (N+/P+)

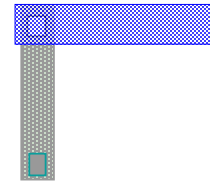
- Is used primarily to alter the type and level of conductivity of semiconductor materials.
- Is used to form bases, emitters, and collectors in bipolar devices, as well as drains and sources in MOS devices.



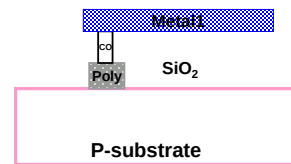
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Contacts

- Create a path, stitching on multiple layers with connections.
- Are all symbolic devices that are defined in the technology file.
- Have attributes that are the contact type, width, and length, and rows and columns.



Top view

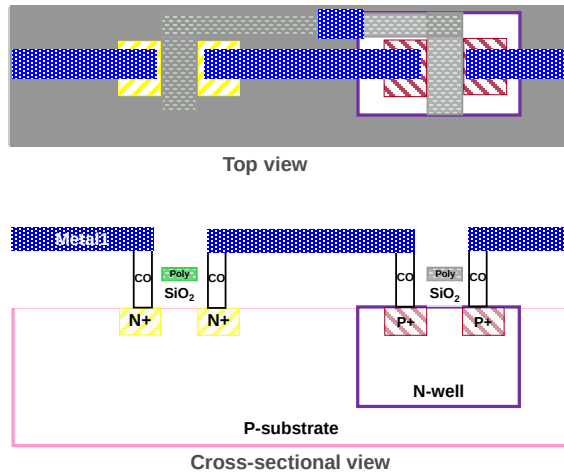


Cross-sectional view

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Metals

- Act as interconnects to the circuits that constitute several layers on ICs, depending on the process, for example, M1 through M7.
- Are usually made of aluminum, copper, or aluminum-copper alloy.



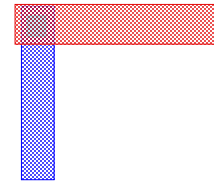
Al is very suitable for this purpose with its very low resistivity and its adhesion compatibility with SiO₂. A disadvantage of Al as the metallization material is its low melting temperature (660 deg C) and the low Al-Si eutectic temperature (577 deg C). These restrict the maximum processing temperature once the Al layer has been deposited.

Vias

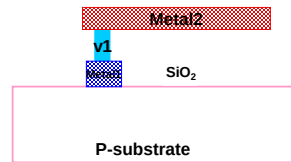
- Are connection points between wires of different types.
- Are used for connecting different metal layers, for example, V1 connects M1 and M2.

Anatomy of a via

- Via coverage (metal2)
- Via cut (v1)
- Via landing (metal1)



Top view

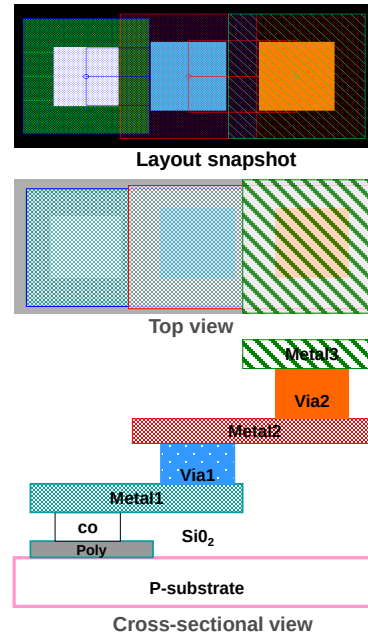


Cross-sectional view

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Conventional Vias

- Are supported mostly in an older process
- Are high in area consumption



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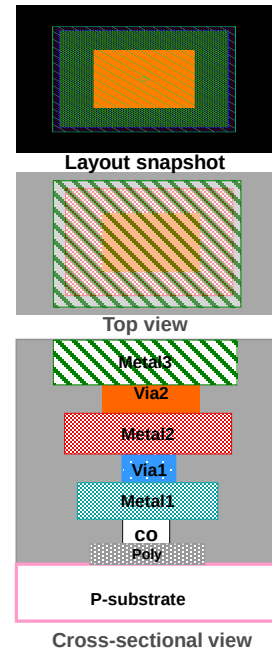
Picture TIIT/TTM (2008)

Via can be classified into two types:

- **Conventional Via (Area consuming).**
- Stacked Via (e.g., For achieving higher routing density, Not all processes support this).

Stacked Vias

- Are supported mostly by the latest processes
- Are efficient in area consumption
- Can be used in high-congestion designs
- Have less resistance than conventional vias



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Picture TIIT/TTM (2008)

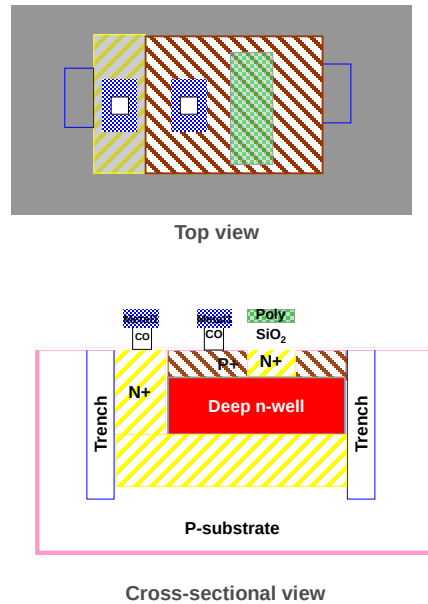
Via can be classified into two types:

- Conventional Via (Area consuming).
- **Stacked Via (e.g., For achieving higher routing density, Not all processes support this).**

Via resistance is less for a stacked via compared to conventional via for the same stack depth.

Trench Isolation

- Shallow trench isolation (STI), also known as [box isolation technique](#), is an integrated-circuit feature that prevents electrical current leakage between adjacent semiconductor device components.
- STI is generally used on CMOS process technology nodes of 250 nanometers and smaller.



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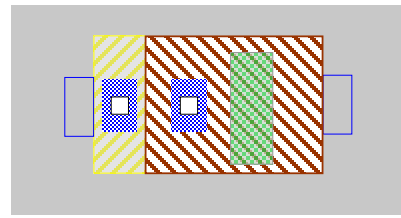
Older CMOS technologies and noNMOSTechnologies commonly use isolation based on [LOCOS](#).^[1]

STI is created early during the [semiconductor device fabrication](#) process, before transistors are formed. The key steps of the STI process involve etching a pattern of trenches in the silicon, depositing one or more [dielectric](#) materials (such as [silicon dioxide](#)) to fill the trenches, and removing the excess dielectric using a technique such as [chemical-mechanical planarization](#).^[1]

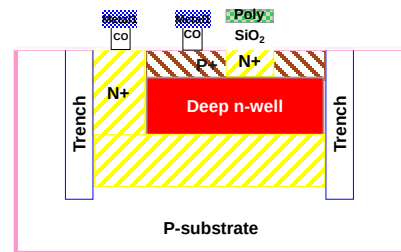
Certain semiconductor fabrication technologies also include deep trench isolation, a related feature often found in [analog](#) integrated circuits.

Deep N-Well (or Buried N-Well)

- Is a n-well layer implanted deeper down in the substrate than the normal n-well.
- Is used mainly to control the substrate leakage of NMOS and to provide better isolation from substrate.



Top view



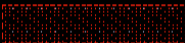
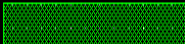
Cross-sectional view

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In mixed signal circuits there is only an imaginary line to separate the Digital and Analog parts. The SUB is common for both. Digital circuits usually operate at higher frequencies so that they switch on and off at a very faster rate. They dump minority charge carriers in the SUB which can affect the analog NMOS behaviors. There is a need to protect analog NMOS transistors from this sub noise so there is a concept of Deep N Wells.

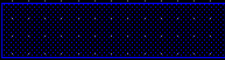
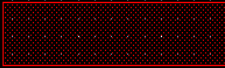
Process Layer Representation

Layer name	Color		GPDK90 colors
	Fill	Line	
Psub	Black	Rose	 P substrate
N-well	Black	Violet	 nwell
Oxide	Brown pattern	Brown dashed	 oxide
Poly	Green cross fill	Green	 poly
PIMP	Black diagonal pattern	Brown	 P implant
NIMP	Yellow diagonal pattern	Yellow	 N implant



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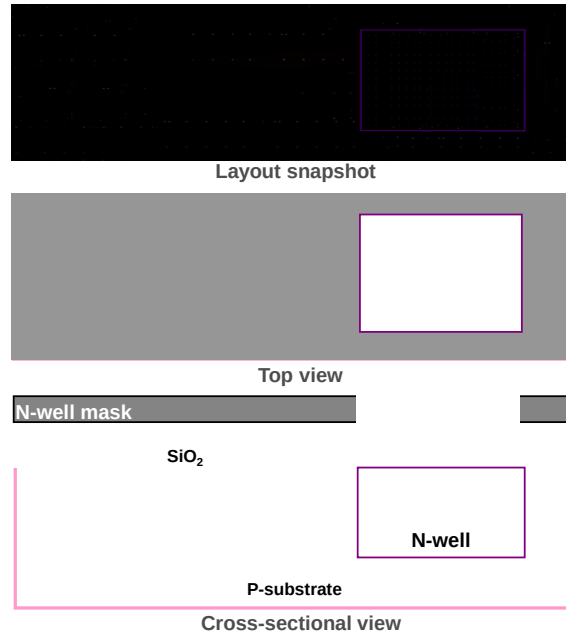
Process Layer Representation (continued)

Layer name	Color		GPDK90 colors
	Fill	Line	
Contact	White	White	 Contact
Metal1	Blue pattern	Blue	 Metal1
Via1	Sky blue	Sky blue	 Via1
Metal2	Red pattern	Red	 Metal2
Via2	Orange	Orange	 Via2



This page does not contain notes.

Design Layer Correspondence: N-Well



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This slide is done according to gpdk090 layers given in AMS5.1 Kit.

Sequence of Fabrication of CMOS

Epitaxy: Single crystal film grown on silicon

- A p-type single crystal film is grown on the surface of the substrate by:
 - Subjecting the substrate to high temperature and a source of dopant material.

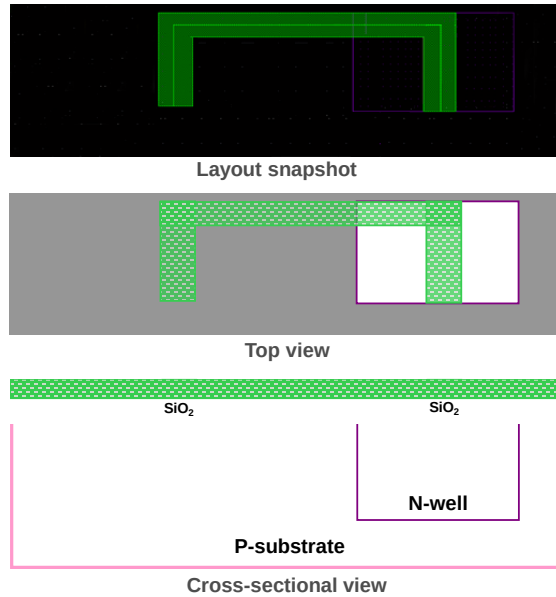
N-well Formation

- The first mask defines the N-well regions
- N-wells are formed by ion implantation or diffusion
- PMOS transistors are fabricated in N-well regions

Gate oxide growth

- Controlled gate oxide is grown with thickness.
- The device's threshold voltage is adjusted by:
 - Adding charge at the silicon/oxide interface.

Design Layer Correspondence: Poly

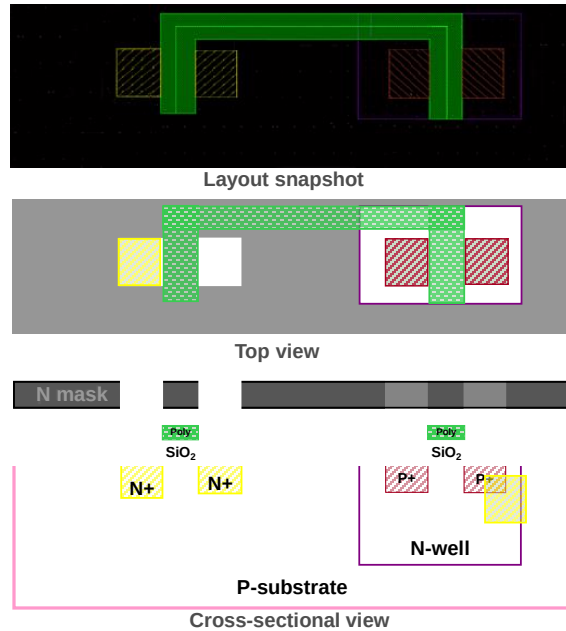


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- A layer of polysilicon is deposited over the entire wafer surface.
- The polysilicon is then patterned by photo-lithography sequence.
- All the MOSFET gates are defined in a single step.

Design Layer Correspondence: Diffusion



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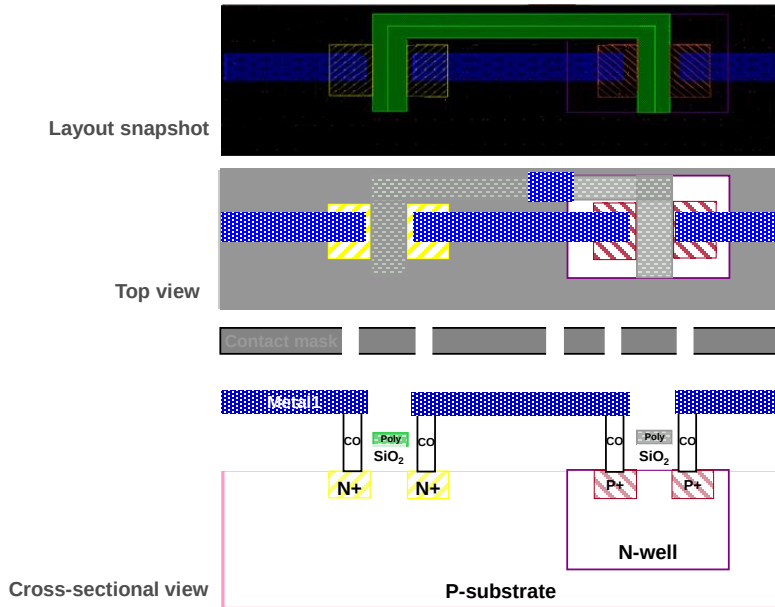


- A boron ion beam creates the p+ source and drain regions.
- The polysilicon serves as a mask to the underlying channel.
 - This is called a self-aligned process.
 - It allows precise placement of the source and drain regions.
- During this process the gate gets doped with p-type impurities
 - The final type (n or p) will depend on which dopant is dominant.

N-well Formation

- PMOS transistors are fabricated in N-well regions.
- The first mask defines the N-well regions.
- N-wells are formed by ion implantation or diffusion.

Design Layer Correspondence: Contact



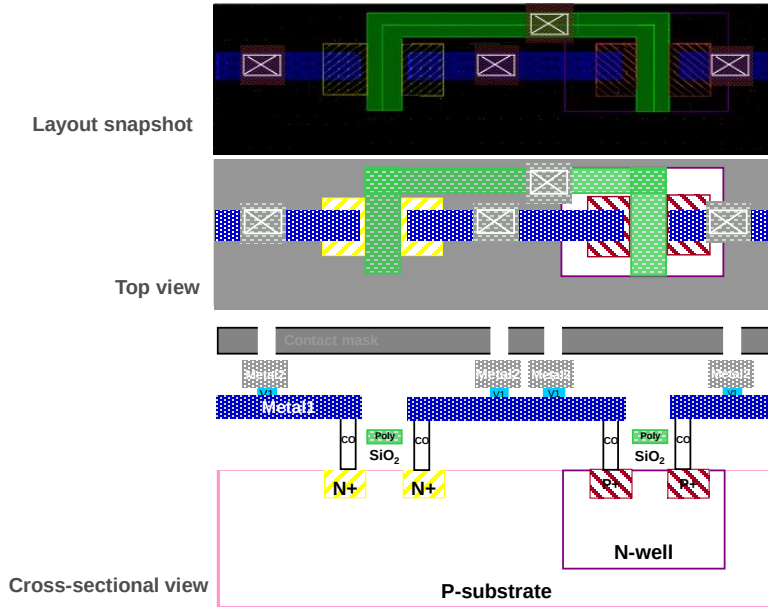
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- The surface of the IC is covered by a layer of Chemical vapor Deposition (CVD) oxide.
 - The oxide is deposited at low temperature to avoid that underlying doped regions will undergo diffusive spreading.
- Contact cuts are defined by etching SiO₂ down to the surface to be contacted.
- These allow metal to contact diffusion and/or polysilicon regions.

A first level of metallization is applied to the wafer surface and selectively etched to produce the interconnects.

Design Layer Correspondence: Metal/Via



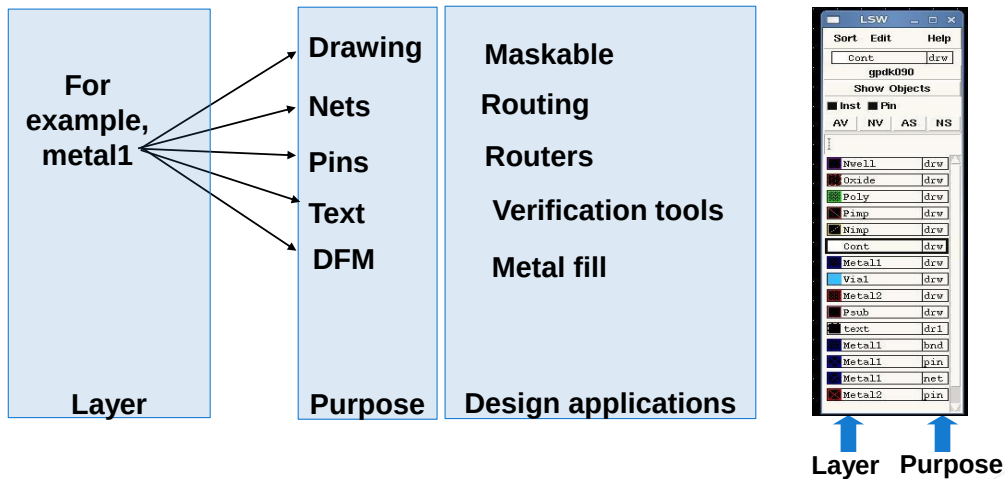
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- Another layer of Low temperature oxide CVD oxide is added.
- Via openings are created.
- metal 2 is deposited and patterned.

Physical and Nonphysical Layers

- *Layer* and *purpose* indicate the applicability of the respective layer.
- Layers are classified with respect to their layer purposes.



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This slide talks about tools.

Determine placement of layout objects

Color coding specifies layers

Layout objects:

- Rectangles
- Polygons
- Arbitrary shapes

Physical and Nonphysical Layers (continued)

- **Drawing layers:** Used to prepare masks.
- **Nets:** Logical signal connections between a set of pins on different instances. After routing, a net consists of routed wires on the routing layers:
 - Interconnects can be drawn with nets layer purpose.
 - CAD tools utilize these layers for routing (connectivity).
- **Pins:** A physical implementation of a terminal, an electrical input or output of a net.
CAD tools use these layers for routing and physical verification.
- **Text:** CAD tools use this information to identify connectivity during physical verification.
- **Design for manufacturing (DFM):** Used for metal fill (to fix density violations) to avoid the effect of chemical mechanical polishing (CMP).



You defined in your technology file with a layer-purpose of *drawing*

Layer-Purpose Pair: One layer and one purpose make up a layer-purpose pair.

Layer-purpose pairs are defined in the technology file. You first define the purpose, then the layer, and then put the purposes and layers together to create layer-purpose pairs.

For example, you have the purposes of *net* and *drawing* defined in the *techPurposes* section of the technology file. Also, the layer *via* is defined in the *techLayers* section of the

technology file. Then in the *techLayerPurposePriorities* section you could create the layer-purpose pair of *via net*, *via drawing*.

Physical and Nonphysical Layers (continued)

Nonphysical layers

- Do *not* contribute to any mask during tapeout.
- Are used by several Virtuoso® products for automation, although they do not contribute to the circuit performance.
- Are shown in the following examples:
 - :text
 - :pin
 - :net
 - :spnet
 - :marker layers

Physical layers

- Are layers that contribute to the mask generation
- Contribute to the actual creation of circuits on silicon
- Are converted to the final geometric data stream (GDS)
- Are shown in the following examples:
 - metal:drw and :fill
 - wells:drw and :fill
 - diffusion:drw
 - vias:drw and :fill
 - Poly-si:drw and :dummy, and so on.



2.1.4

In the technology file you specify the layers used for connectivity, layers that have equivalent connectivity, the layers that cannot overlap, layers that are used during automatic routing and the design rules that apply to those layers. Difference between the physical and non-physical layers.

Display.drf Specifies how you want your layers to appear in the display devices you use.



Subsection
Connectivity

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Connectivity: Physical Structures

Connectivity: An electronic relation between the electrical data and the physical data.

- In mixed-signal flows, connectivity is defined from the schematic and netlist sources. The tools that use this information and provide the automation are Virtuoso XL, Virtuoso Chip Assembly Router, and Virtuoso Preview.
- Connectivity plays a major role through out the design cycle. It is being used and checked during several phases of the design cycle, for example:
 - Schematic: check and save.
 - Layout XL: show unconnected nets.
 - Virtuoso Chip Assembly Router: routing the unconnected nets.
 - Verification: LVS.



Layer definitions (layer numbers, purposes, priorities, and connectivity)

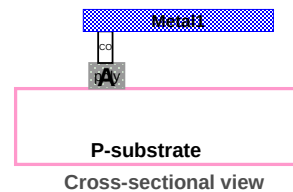
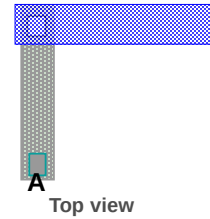
VXL : Virtuoso XL,

VCAR : Virtuoso chip assembly router

LVS : layout versus schematics

Connectivity: Metal-Contact-Poly

Connectivity of the designs (physical structures and layers) are understood through the connectivity components of the respective tools through which the information is transferred throughout the design cycle.



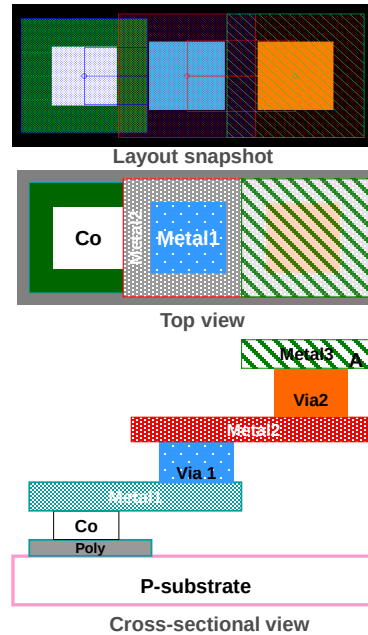
Flow chart for connectivity components for schematic

Connectivity of schematic physical structures: While building the schematics: The connectivity of the schematics, is realised through the in-build capability of virtuoso schematic editor(during check & save).

During netlister, the CDF (component description format) of all the primitive components are used to realise the connectivity of the design.

Connectivity: Metal-Via-Metal-Contact-Poly

In the layout environment, connectivity of the nets can traverse to any metal through vias that are defined in the technology file.



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Picture TIIT/TTM (2008)

Via can be classified into two types:

- **Conventional Via (Area consuming).**
- Stacked Via (e.g., For achieving higher routing density, Not all processes support this).

A → netname.

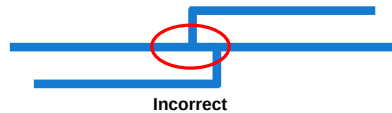
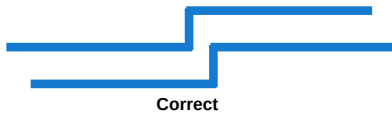
Connectivity: Violations

Connectivity issues: With the connectivity components (for example, pin, net), you can use tools to identify opens, shorts, unconnected pins, and unrouted nets early in the design cycle through violations (for example, Virtuoso XL).

- Opens: Open circuits occur when continuity is not maintained over the entire length of a layout net.



- Shorts: A short circuit occurs when two nets, which should be isolated from each other, are connected.



- Identify electrical problems that can be caused by floating wells or substrate due to insufficient well ties or substrate ties.
- To identify these problems, you define connectivity rules in the rules files. For connectivity rules to work correctly, you must have text in your layout defining the power and ground signals.
- Opens number of layout nets > number of source nets.

Connectivity: Violations (continued)

- Unconnected wires: geometry antenna (dangling wires).
- Unconnected pins: pins that are not connected to any other objects.

Manufacturability issues: Potential problems with the manufacturing process are usually checked during the design rule checking (DRC) process. Even if the connectivity exists, the physical size might not be manufacturable. More information about this topic follows.



Short number of layout nets < number of source nets



Subsection

Design Rules

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Design Rules

- Process-specific design rules must be followed when drawing layouts to avoid manufacturing defects during the fabrication of an IC.
- Process design rules are the minimum allowable drawing dimensions.
- Design rules affect the X and Y dimensions of the layout – *not* the depth or vertical dimensions:
 - The smallest geometry and spacing that can accurately be fabricated.
 - The smallest geometry and spacing that maintains electrical isolation and/or functionality.

We know

What we draw is *not* what we get in silicon.



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- Need of Design Rules:
- Manufacturing problems
- Photoresist shrinkage, tearing.
- Variations in material deposition.
- Variations in temperature.
- Variations in oxide thickness.
- Impurities.
- Variations between lots.
- Variations across a wafer.
- Transistor problems
- Variations in threshold voltage:
 - oxide thickness;
 - ion implantation;
 - poly variations.
- Changes in source/drain diffusion overlap.
- Variations in substrate.

Why Do We Need Design Rules?

- Design rules specify geometric constraints on the layout artwork.
- Design rules represent the best compromise between *performance* and *yield*:
 - More *conservative* rules increase *yield*.
 - More *aggressive* rules increase *performance*.
- Design rules represent a *tolerance* that ensures *high probability* of correct fabrication.

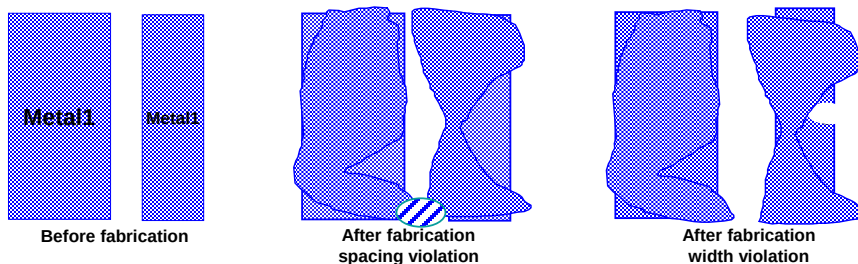
Note: Design rules are *not hard boundaries* between correct and incorrect fabrication.



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Design-Rule Violation

Violating a design rule might result in a nonfunctional circuit or low yield.



As fabrication process is not perfect, we need to use minimum microns to avoid violations.

Types of Design Rules

Scalable design rules: lambda parameter

- Based on scalable coarse grid: L (lambda)
- **Idea:** Reduce L value for each new process, but keep rules the same:
 - Key advantage: portable layout
 - Key disadvantage: not everything scales the same
- Not used in real-world situations

Absolute dimensions (micron rules)

- Are based on absolute distances (for example, $0.75\ \mu\text{m}$)
- Are tuned to a specific process (details usually proprietary)
- Are complex, especially for deep submicron
- Have layouts that are not portable

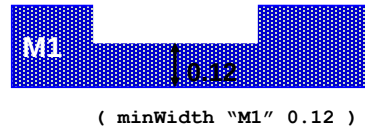


Lambda and micron rules: now in industry we use micron rule.

Typical Design Rules

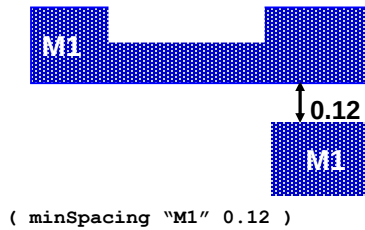
Minimum width (minWidth)

Specifies the minimum width of individual shapes on a single layer. Width is the difference between inside facing edges of individual shapes.



Minimum spacing (minSpacing)

Defines the minimum spacing between objects on a single layer. It can also be viewed as the shortest distance between two edges of different shapes of the same layer.

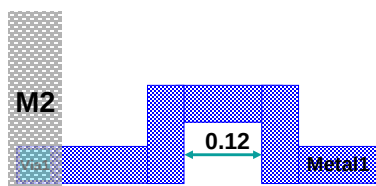


Process design rules are Contact, Minimum Dimensions, Minimum Spacing and Minimum Enclosure.

Typical Design Rules (continued)

Minimum same net spacing (minNotch)

Specifies the minimum spacing rule for objects on the same net, including defining the minimum notch on a single-layer, merged object.



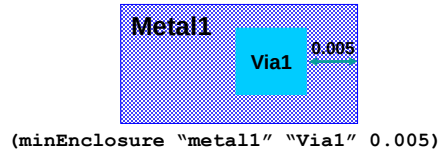
```
(minNotch "metal1" 0.12)
```

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Typical Design Rules (continued)

Minimum enclosure (minEnclosure)

- Specifies the enclosure of shapes on the second layer by shapes on the first layer. The rule implies that the second layer is fully enclosed by the first one.
- The enclosure is the distance between inside-facing edges of shapes on the first layer to outside-facing edges of shapes on the second layer.

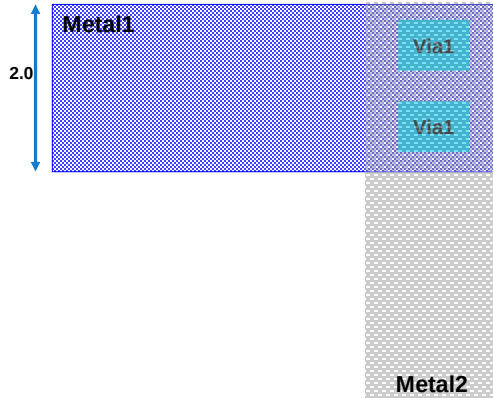


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Typical Design Rules (continued)

Minimum cut (minimumCut)

- Defines the minimum number of cuts a via must have when it is on a wide wire or pin whose width is greater than the width specified in the rule.
- Applies to all vias containing the cut layer specified in the rule.



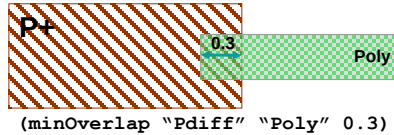
```
(minimumCut "metal1" 2.0000 WIDTH 2.0000")
```

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Typical Design Rules (continued)

Minimum overlap (minOverlap)

- Specifies the overlap of shapes on one layer by shapes on the other layer. (This is different from the *minEnclosure* rule that implies that the object on the second layer is fully enclosed in the first one.)
- The overlap is the distance between an inside-facing edge of a shape on one layer to an inside-facing edge of a shape on the other layer.



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Physical Structure of Devices

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Module Objectives

In this module, you learn how to

- Recognize top and cross-sectional views of devices.
- Identify to which device these views belong.

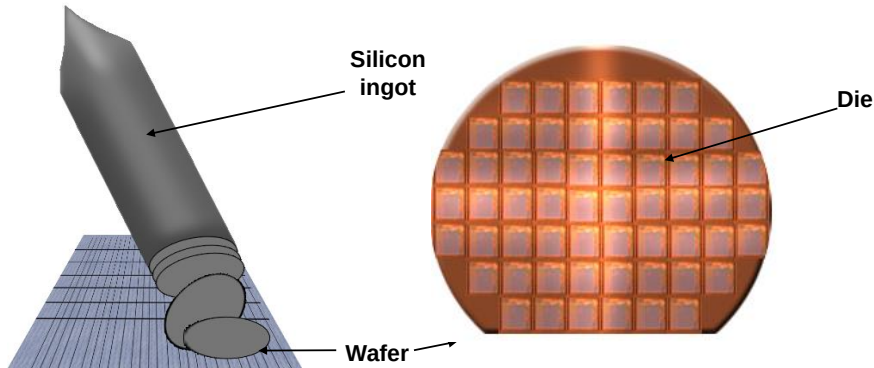


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Terminology: Wafer

Wafer: A thin slice of semiconductor material, such as a silicon crystal, used in the fabrication of integrated circuits.

- Disk of silicon, 4 in. to 18 in. in diameter, < 1 mm thick, cut from ingots of single-crystal silicon, which are grown from a pot of melted silicon.



Source of the picture: TTM 2008.

The pure semiconductor silicon or germanium has four valence electrons. In silicon, the most common dopants are IUPAC group 13 (commonly known as *group III*) and group 15 (commonly known as *group V*) elements. Group 13 elements all contain three valence electrons, causing them to function as acceptors when used to dope silicon. Group 15 elements have five valence electrons, which allows them to act as a donor. Therefore, a silicon crystal doped with boron creates a p-type semiconductor whereas one doped with phosphorus results in an n-type material.

Terminology: Chips, Substrates, and Wells

Chips and dies

- A die is one of the repeated ICs on a wafer.
- A chip is a packaged die.

Substrates and bodies

- Substrate is a thin slice of semiconductor material upon which semiconductor devices are fabricated.
- Substrate regions are fundamentally doped with n-type and/or p-type impurities.

N-wells and P-wells

- A well acts as an intermediate substrate for the respective MOS devices of semiconductor material. These wells are formed with diffusion or ion implantation.
- Wells are doped with n-type or p-type impurities.



Well Proximity requirements have changed. Well Proximity measurements now must take into account the case where well edges are an acceptable distance away from the channel, but the corners are not. This is in accordance with the new BSIM 4.5 MOSFET models.



Subsection

Substrate and Wells

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Substrate and Wells

Substrate: The base layer in integrated circuit fabrication on which device layers are diffused to form a physical structure.

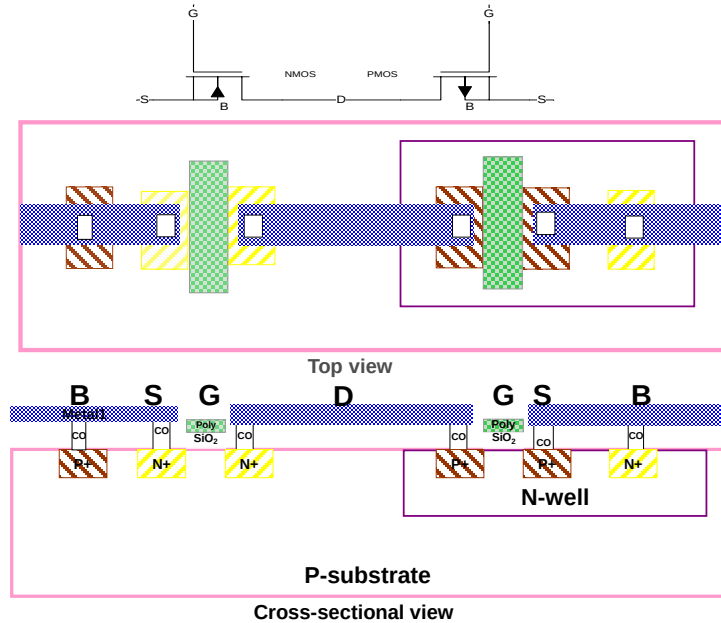
Various CMOS fabrication processes with respect to the substrate and well configuration are as follows:

- P-well process
- Twin-well process
- Shallow trench isolation
- Silicon on insulator (SOI)



If the substrate is P-type then the device built on that substrate is N-type. This is because a reversed biased diode is formed between the substrate and device, which stops the flow of leakage currents into the device. Hence avoids the formation of parasitic structures.

CMOS Fabrication Process



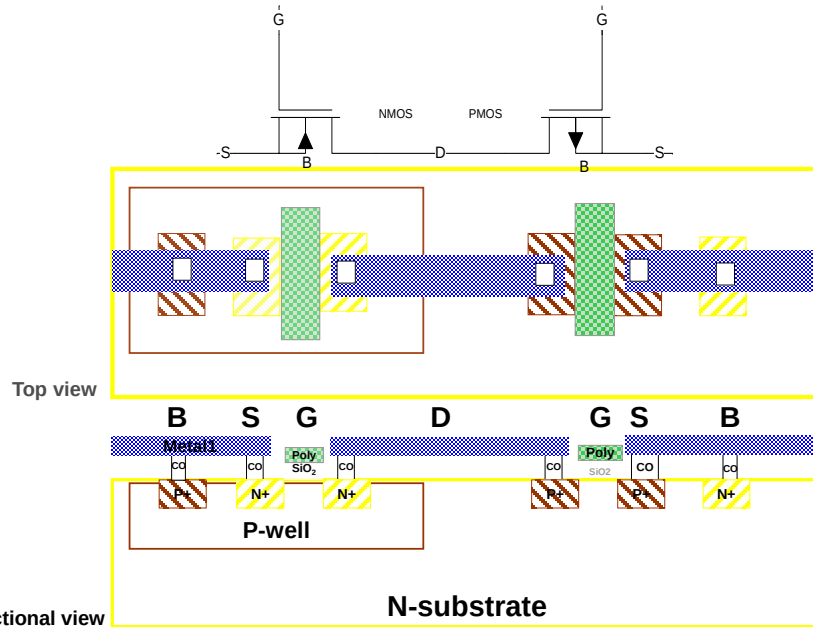
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Picture Source :TTM copyright 2008

- CMOS accommodates both nMOS and pMOS devices, special regions must be created in which the semiconductor type is opposite to the substrate type.
- These regions are called wells or tubs. An N-well is created in a p-type substrate or, alternatively, a P-well is created in an n-type substrate.

P-well Process



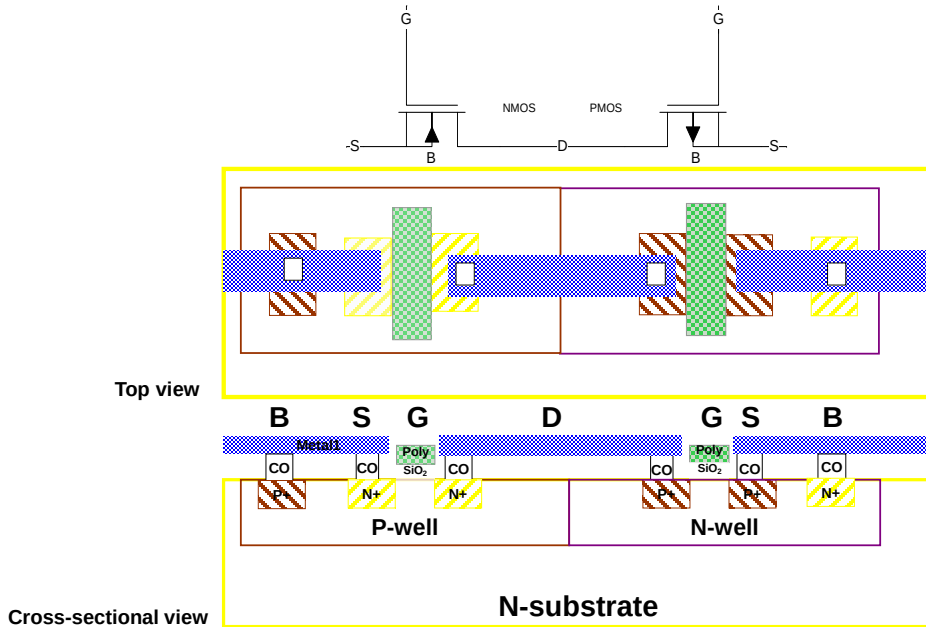
Similar to N-well process except a P-well is implanted rather than an N-well.

In general, p-devices are lower gain than n-devices.

Therefore, P-well process naturally moderate the differences.

- NMOS devices are built on an implanted P-well.
- PMOS devices are built on the substrate.
- P-well process moderates the difference between the p- and the n-transistors since the P devices reside in the native substrate.
- Advantages: better balance between p- and n-transistors

Twin Tub Process



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This technology provides the basis for separate optimization of the nMOS and pMOS transistors, thus making it possible for threshold voltage, body effect and the channel transconductance of both types of transistors to be tuned independently. Generally, the starting material is a n+ or p+ substrate, with a lightly doped epitaxial layer on top. This epitaxial layer provides the actual substrate on which the N-well and the P-well are formed. Since two independent doping steps are performed for the creation of the well regions, the dopant concentrations can be carefully optimized to produce the desired device characteristics.

In the conventional N-well CMOS process, the doping density of the well region is typically about one order of magnitude higher than the substrate, which, among other effects, results in unbalanced drain parasitics. The twin-tub process (Fig. 2.12) also avoids this problem.

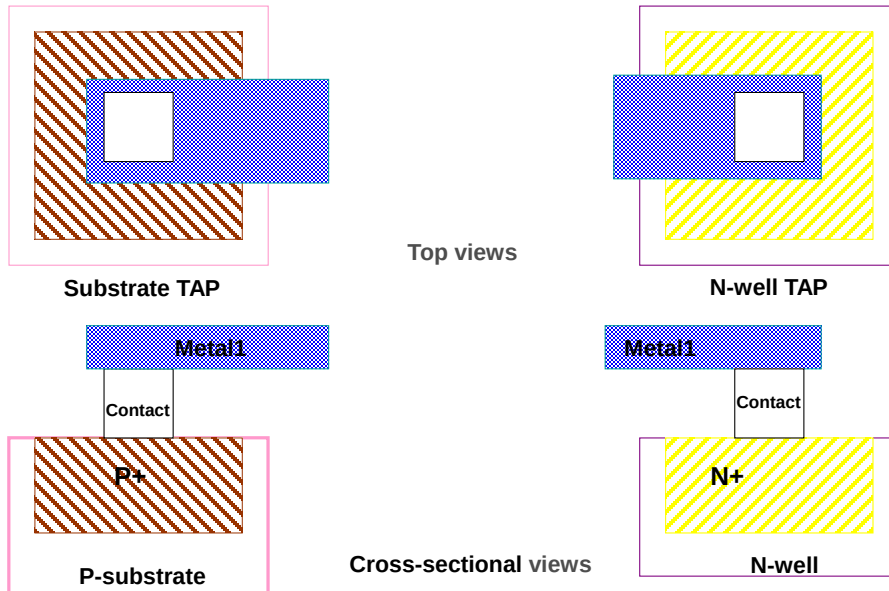
Other advantages of using the twin-tub process is that a self-aligned twin-tub structure can be achieved using only one mask.

Epitaxy is high purity Silicon grown with accurately determined dopant concentrations.

Both type of substrate contacts are required in this process.

- n+ or p+ substrate plus a lightly doped epi-layer (latchup prevention).
- Wells for the n-transistors and p-transistors.
- Advantages: Two independent doping steps are performed for the creation of the well regions, the dopant concentrations can be carefully optimized to produce the desired device characteristics.

Substrate and Well Taps



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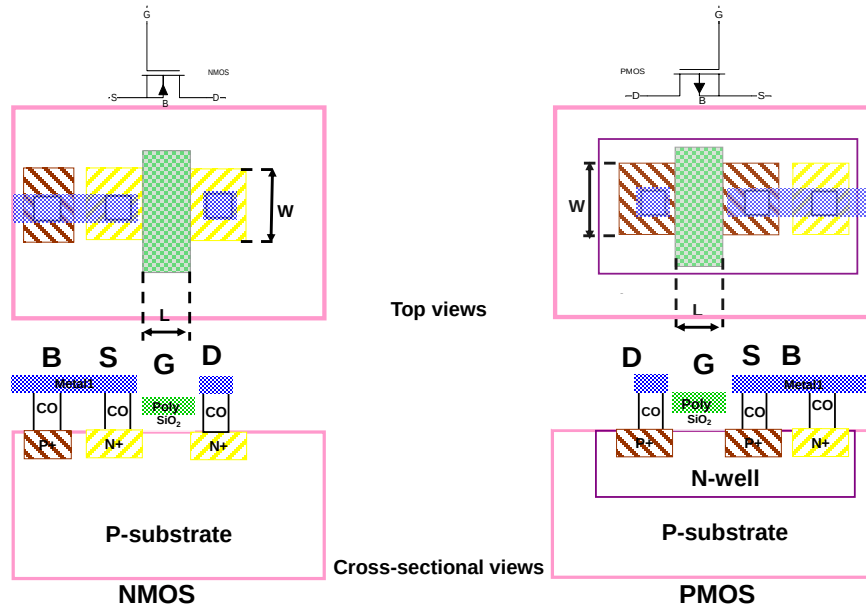
Subsection

Integrated Devices

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MOS Transistors



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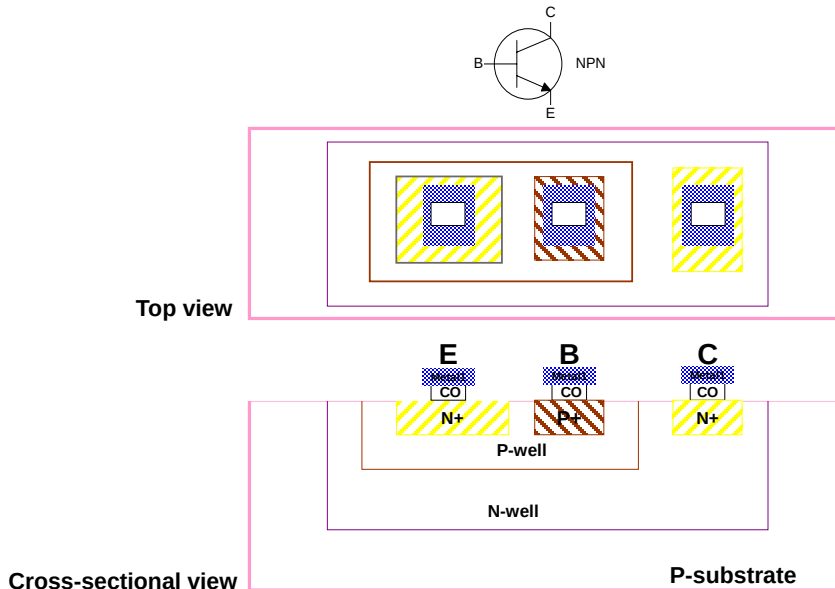
The most basic element in the design of a large-scale integrated circuit is the transistor. For the processes we will discuss, the type of transistor available is the Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET). These transistors are formed as a “sandwich” consisting of a semiconductor layer, usually a slice, or wafer, from a single crystal of silicon; a layer of silicon dioxide (the oxide) and a layer of metal. These layers are patterned in a manner which permits transistors to be formed in the semiconductor material (the “substrate”).

Actually, the transistors which we will use do not use metal for their gate regions, but instead use polycrystalline silicon (poly). Polysilicon gate FET's have replaced virtually all of the older devices using metal gates in large scale integrated circuits. (Both metal and polysilicon FETs are sometimes referred to as IGFET's --- insulated gate field effect transistors, since the silicon dioxide under the gate is an insulator. We will still continue to use the term MOSFET to refer to polysilicon gate FETs.)

The transistor consists of three regions, labeled the “source,” the “gate,” and the “drain.” The area labeled as the gate region is actually a “sandwich” consisting of the underlying substrate material, which is a single crystal of semiconductor material (usually silicon); a thin insulating layer (usually silicon dioxide); and an upper metal layer. Electrical charge, or current, can flow from the source to the drain depending on the charge applied to the gate region. The semiconductor material in the source and drain region are “doped” with a different type of material than in the region under the gate, so an NPN or PNP type structure exists between the source and drain region of a MOSFET.

- Add “source” and “drain” terminals to MOS capacitor.
- Transistor types
 - NMOS: p-type substrate, n^+ source/drain.
 - PMOS: N-well in p-type substrate, p^+ source/drain

Bipolar Transistors: NPN



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Picture Source: TTM

An NPN transistor can be considered as two diodes with a shared anode region. The electrons in the base are called *minority carriers* because the base is doped p-type which would make holes the *majority carrier* in the base.

A BJT consists of three differently [doped](#) semiconductor regions, the *emitter* region, the *base* region and the *collector* region. These regions are, respectively, *p* type, *n* type and *p* type in a [PNP](#), and *n* type, *p* type and *n* type in a [NPN transistor](#). Each semiconductor region is connected to a terminal, appropriately labeled: *emitter* (E), *base* (B) and *collector* (C).

The *base* is physically located between the *emitter* and the *collector* and is made from lightly doped, high resistivity material. The collector surrounds the emitter region, making it almost impossible for the electrons injected into the base region to escape being collected, thus making the resulting value of α very close to unity, and so, giving the transistor a large β . A cross section view of a BJT indicates that the collector–base junction has a much larger area than the emitter–base junction.

The bipolar junction transistor, unlike other transistors, is usually not a symmetrical device. This means that interchanging the collector and the emitter makes the transistor leave the forward active mode and start to operate in reverse mode. Because the transistor's internal structure is usually optimized to forward-mode operation, interchanging the collector and the emitter makes the values of α and β in reverse operation much smaller than those found in forward operation; often the α of the reverse mode is lower than 0.5. The lack of symmetry is primarily due to the doping ratios of the emitter and the collector. The emitter is heavily doped, while the collector is lightly doped, allowing a large reverse bias voltage to be applied before the collector–base junction breaks down. The collector–base junction is reverse biased in normal operation. The reason the emitter is heavily doped is to increase the emitter injection efficiency: the ratio of carriers injected by the emitter to those injected by the base. For high current gain, most of the carriers injected into the emitter–base junction must come from the emitter.

The low-performance "lateral" bipolar transistors sometimes used in [CMOS](#) processes are sometimes designed symmetrically, that is, with no difference between forward and backward operation.

Small changes in the voltage applied across the base–emitter terminals causes the current that flows between the *emitter* and the *collector* to change significantly. This effect can be used to amplify the input voltage or current. BJTs can be thought of as voltage-controlled [current sources](#), but are more simply characterized as current-controlled current sources, or current amplifiers, due to the low impedance at the base.

- The bipolar junction transistor (BJT) derives its name from the presence of two charge carriers - electrons and holes.
- BJT structure contain two PN junctions with three terminals.

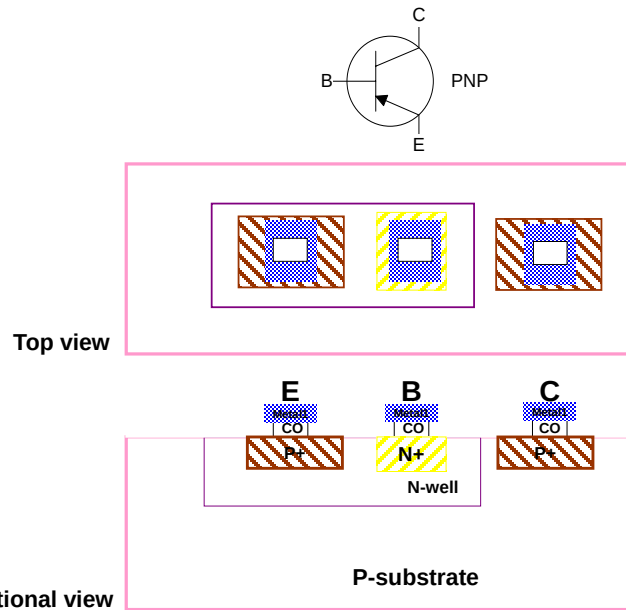
PNP

Emitter, Base and Collector.

NPN

Emitter, Base and Collector.

Bipolar Transistors: PNP



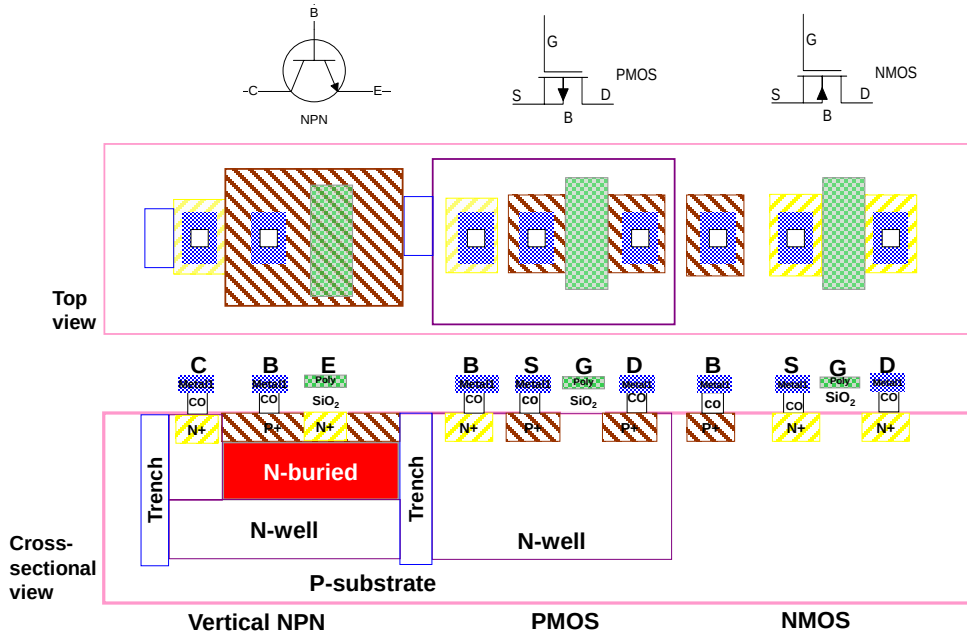
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BiCMOS

- BiCMOS processes are usually based on CMOS process flows, augmented by the addition of bipolar transistors, high-sheet poly resistors, and other special devices.
- BiCMOS gates can be an efficient way of speeding up VLSI circuits.
- A known deficiency of MOS technology is that it has limited load-driving capability due to limited current sourcing and sinking abilities of PMOS and NMOS transistors.
- Bipolar transistors have:
 - Higher gain
 - Better noise margins and high-frequency characteristics
- Example applications of BiCMOS include:
 - Analog I/O and driver circuits
 - High-performance microprocessors



This technology has commercial application in amplifier and discrete component logic design. More recently it has become the technology of choice for power electronics products such as voltage regulators.

BiCMOS (continued)

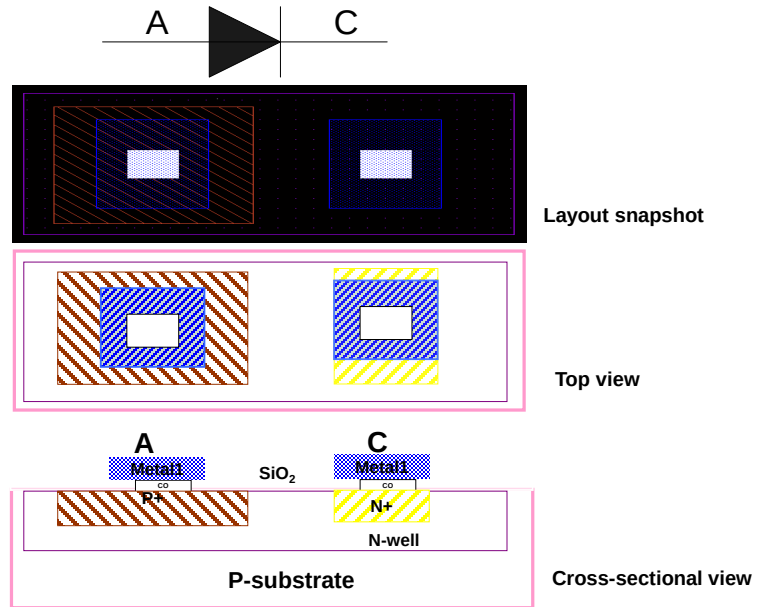
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Picture: Cadence

- Bipolar transistors consume more current (and therefore dissipate more power), but which switch faster.
- It is generally recognized that CMOS remains the backbone for building the digital portion of the systems, and bipolar devices are used primarily for the rf and analog functions.
- **(BiCMOS)** is a type of integrated circuit that uses both bipolar and CMOS technologies. The logic gates are primarily made of CMOS, but their output stages use bipolar junction transistors (BJTs).
- The development of BiCMOS for digital logic applications was to combine the characteristics of high-speed bipolar and low-power CMOS on the same chip integrating bipolar and metal–oxide–semiconductor (MOS) transistors into a single device proved difficult and uneconomical

Integrated Devices: Diode



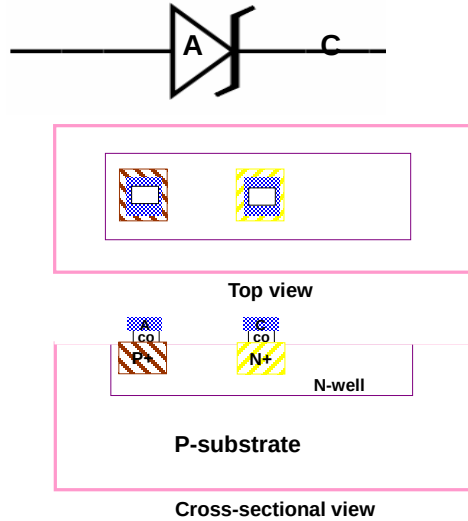
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From gpdk090 library from AMS5.1 kit.

Integrated Devices: Zener Diode



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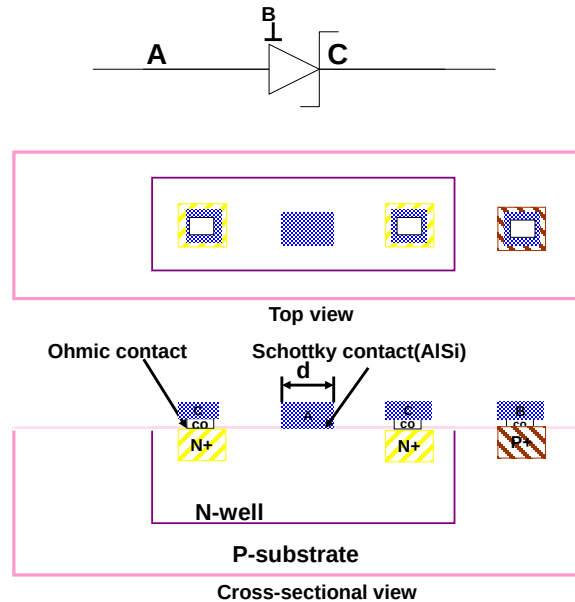
Source of the views: Art of Analog Layout (page no350)

Zener diode, is generally poorly suited for very high-speed I/O interfaces because the lowest capacitance of existing devices is about 30 pF. This capacitance is too high to pass high-frequency signals without significant distortion. This distortion results in unreliable detection of the signals and increased high-frequency roll-off.

A is Anode terminal

C is Cathode terminal

Integrated Devices: Schottky Diode



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Source of the views: Art of Analog Layout (page no354)

Schottky diode current

The current across a metal-semiconductor junction is mainly due to majority carriers. Three distinctly different mechanisms exist: diffusion of carriers from the semiconductor into the metal, thermionic emission of carriers across the Schottky barrier and quantum-mechanical tunneling through the barrier. The diffusion theory assumes that the driving force is distributed over the length of the depletion layer. The thermionic emission theory on the other hand postulates that only energetic carriers, those, which have an energy equal to or larger than the conduction band energy at the metal-semiconductor interface, contribute to the current flow. Quantum-mechanical tunneling through the barrier takes into account the wave-nature of the electrons, allowing them to penetrate through thin barriers. In a given junction, a combination of all three mechanisms could exist. However, typically one finds that only one limits the current, making it the dominant current mechanism.

Schottky Diode

- A junction between metal and silicon forms another type of rectifying diode called schottky diode.
- However, a junction between metal and P+ silicon forms a regular connection, a resistive or ohmic contact, instead of a diode.



Subsection

Passive Elements on IC

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Resistor on Chip

- Resistors provide specific and controlled amounts of electrical resistance. They are useful in a variety of applications, including amplifiers, voltage regulators, and so forth.
- Most processes offer a choice of several resistor materials. Some are better suited for fabricating high-value resistors and others for fabricating low-value ones.
- Following are some types of resistors used in ICs:
 - N-well
 - Diffusion
 - Metal
 - Poly



Thickness of given material is defined when the process is defined

Resistivity of each material is known

Therefore, L and W are the only variables

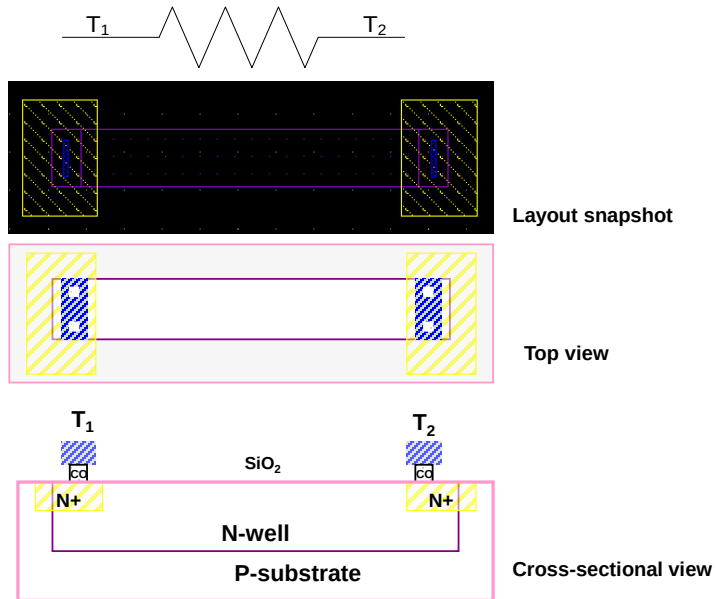
Resistance quoted in Ω/Sq

Material can therefore be measured in “Squares”

Permittivity: $\text{Si} = 1.035 \text{ [pF/cm]}$

- Permeability: $\mu\text{Si} = 4 \text{ [nH/cm]}$
- Resistivity: $= q(p\mu_p + n\mu_n)$

N-well Resistors

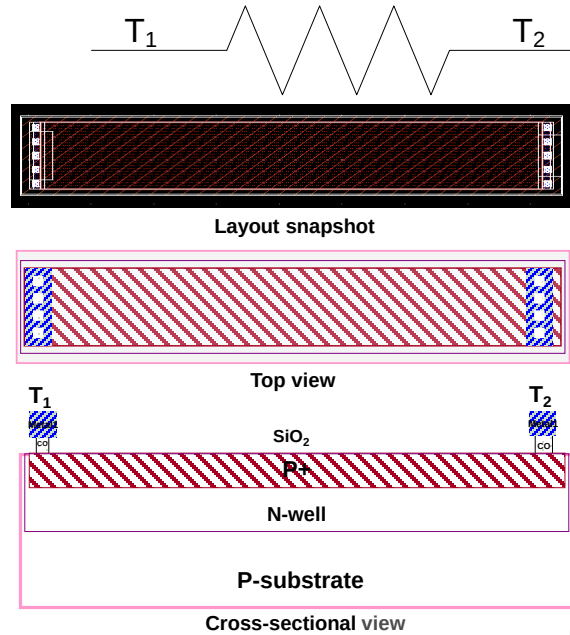


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Diffusion Resistors: P-type

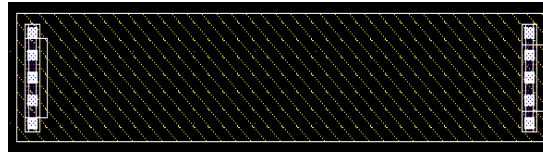
P-type resistor

- Constructed upon n-well
- Low sheet resistance

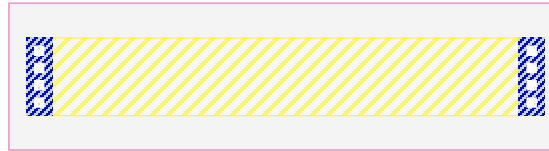


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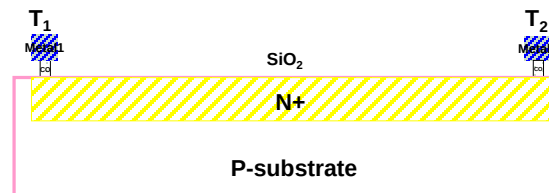
Diffusion Resistors: N-type



Layout snapshot



Top view



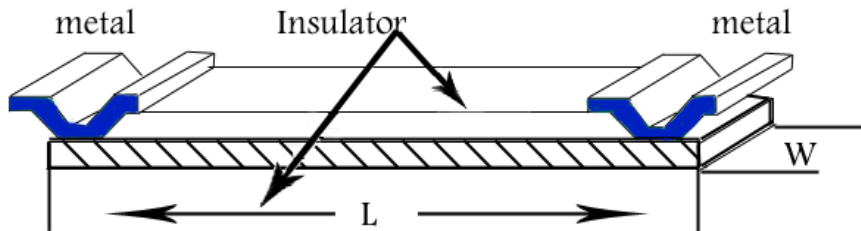
Cross-sectional view



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Integrated Resistors

- Integrated resistors are made of thin strips of resistive layer.
- A resistive strip is contacted at the two terminals by ohmic contacts (metal that is p+ or n+). Insulation from the surroundings is made by oxide layers.



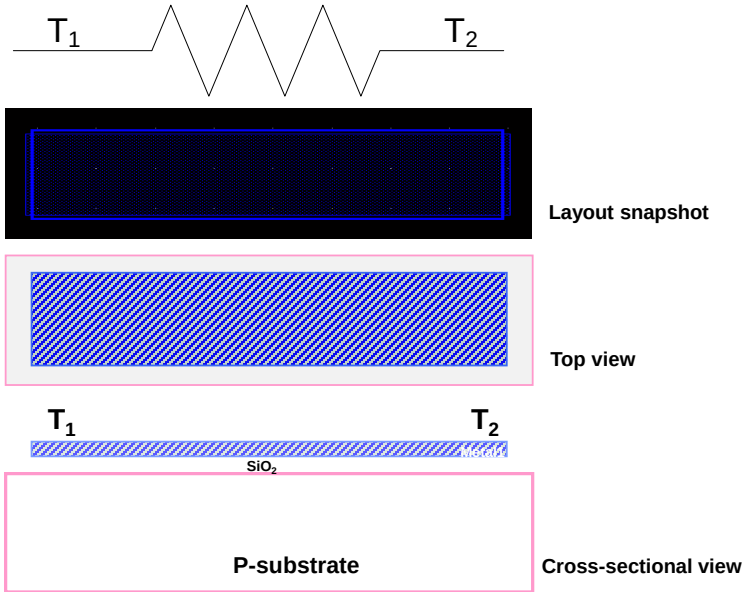
$$R = 2R_{cont} + (L/W) R_{\square} = 2R_{cont} + (\rho/t) (L/W)$$

ρ_s = sheet resistance

Picture Source: TTM.

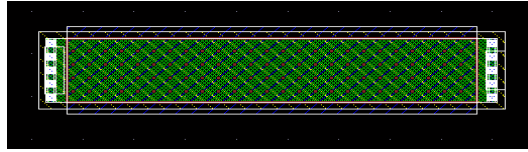
Sheet is given the technology file provided by the foundry.

Metal Resistor: Metal1



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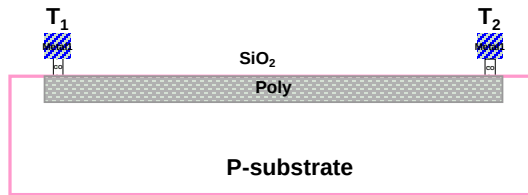
Poly Resistor



Layout snapshot



Top view

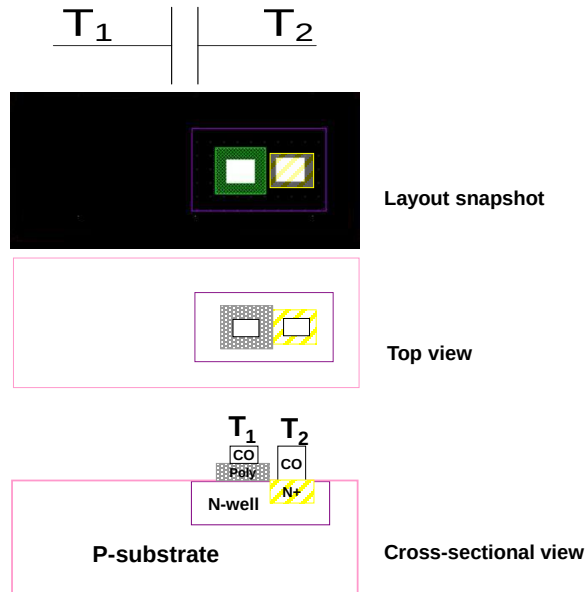


Cross-sectional view



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N-well Capacitor



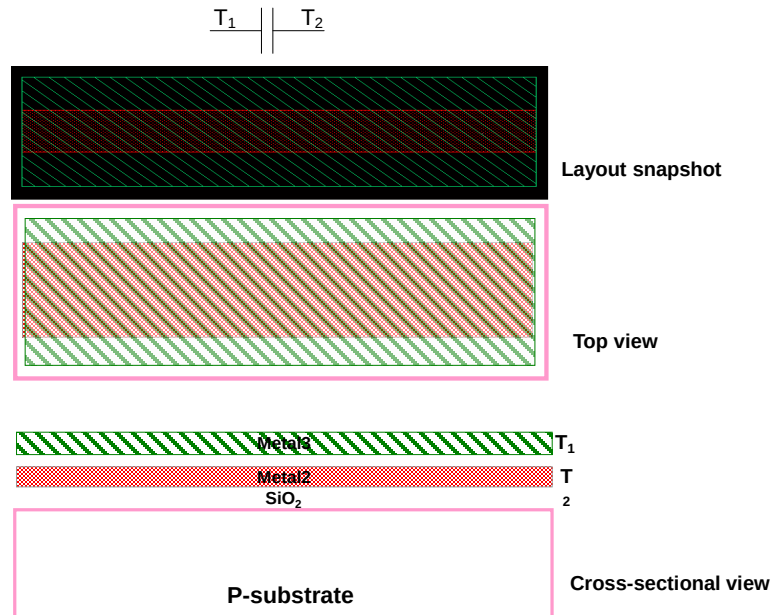
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Drawbacks:

- Large C_{junc} of Bottom Plate.
- Large series R .
- Nonlinearity.
 - One technique for forming a capacitor consists of depositing one polysilicon layer, oxidizing this to form a thin dielectric, and depositing second polysilicon layer to complete the capacitor.
 - Variations in inter-layer dielectric cause variations in capacitance.
 - Depletion Capacitance is a function of Voltage.

Metal-Insulator-Metal Capacitor



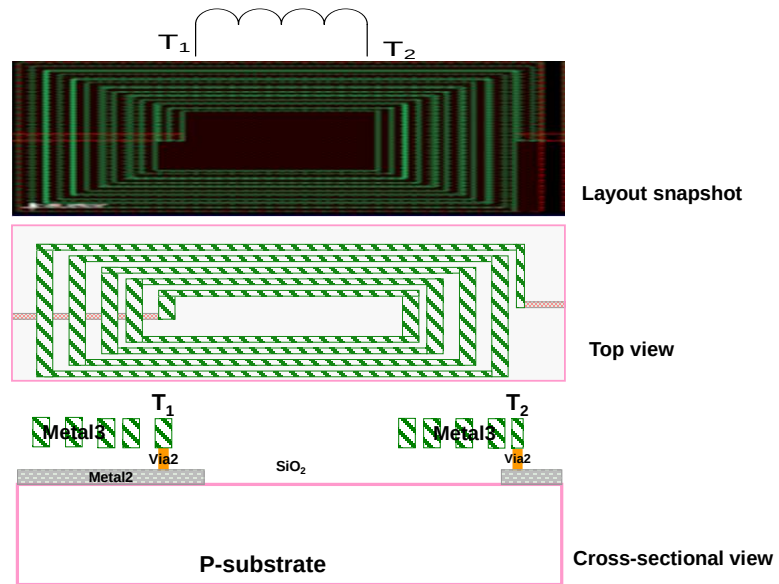
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Junction capacitors customarily employ one of two competing styles of layout.

1. The Plate capacitor, maximizes junction area.
2. Comb capacitor, maximizes junction periphery.
 - Oxide can be thermally grown,
 - Top plate can be Poly or Metal.
 - Bottom plate can be Metal, Poly or Diffusion layers.
 - Prevention of misalignment errors:
 - The upper plate is made smaller than the lower one.
 - The area of the upper plate is assumed to be that of the capacitor.

Inductor Layout



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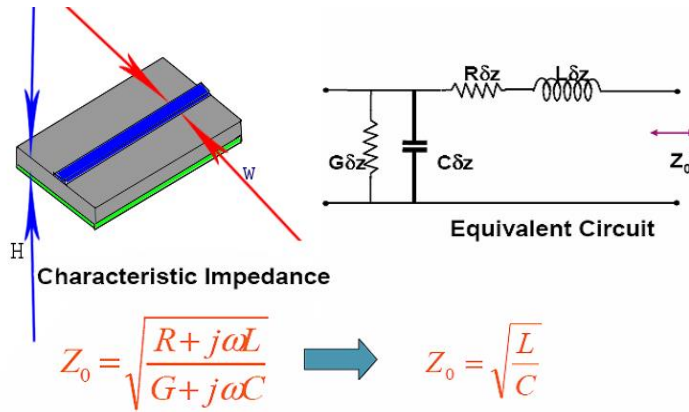
Picture Source: TTM

From gpdk090 library

Lossless Transmission Lines

Lossless transmission lines propagate signals with no distortion or attenuation.

This requires $R = G = 0$, yielding the following:



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Picture Source : TTM

Z_0 : characteristic impedance of the line.

R : conductor resistance.

L : conductor Inductance.

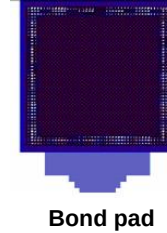
G : dielectric conductance.

C : capacitance.

j : Complex Number

Bond Pads

- Enable the connection of the IC to the external world.
- Are pads through which package pins are connected. These are just like an interface between package to internal core.
- Consist of all metal layers and passivation openings.
 - In layout stage, the opening of passivation is shown with the pad layer.
 - In some technologies, bondpad can be only the top metal layer and pad layer.
- Require, in most cases, electrostatic discharge (ESD) protection circuitry.



Source: cadence.com (white papers)

Bonding Pad Height: Height of the upper face of the die, where the bonding pads are located. The reference height 0 is the bottom face of the package body. This height can be different from the height of the bond pad site on the pin, or on the die attach.

Bondpad Coord (Yb): If the bondpad is on the bottom or top side of the die cavity, then *Bondpad Coord* is the x-coordinate of the center of the bondpad. If the bondpad is on the left or right side of the die cavity, then *Bondpad Coord* is the y-coordinate of the center of the bondpad. The origin of the coordinate system is the center of the die cavity.

Bondpad Length (Lb): Length of the end segment of the pin with bond wires connecting to it.

Bondpad Width (Wb): Width of the end segment of the pin with bond wires connecting to it.

The leadframe file of a 14-pin SOIC package is as follows:

```
bondPad(1) = left -0.2 0.2 1.0 0 pinPad(1) = bottom -3.81 0.49 1.0 1 0 bondPad(2) = left -0.6 0.2 0.49 pinPad(2) =
bottom -2.54 0.49 1.0 2 0 bondPad(3) = bottom -0.6 0.2 0.2 pinPad(3) = bottom -1.27 0.49 0.2 3 0 bondPad(4) =
bottom -0.0 0.2 0.2 pinPad(4) = bottom 0.0 0.49 0.2 4 0 bondPad(5) = bottom 0.6 0.2 0.2 pinPad(5) = bottom 1.27
0.49 0.2 5 0 bondPad(6) = right -0.6 0.2 0.49 pinPad(6) = bottom 2.54 0.49 1.0 6 0 bondPad(7) = right -0.2 0.2 1.0
pinPad(7) = bottom 3.81 0.49 1.0 7 0 bondPad(8) = right 0.2 0.2 1.0 pinPad(8) = top 3.81 0.49 1.0 8 0 bondPad(9)
= right 0.6 0.2 0.49 pinPad(9) = top 2.54 0.49 1.0 9 bondPad(10) = top 0.6 0.2 0.2 pinPad(10) = top 1.27 0.49 0.2
10 0 bondPad(11) = top 0.0 0.2 0.2 pinPad(11) = top 0.0 0.49 0.2 11 0 bondPad(12) = top -0.6 0.2 0.2 pinPad(12)
= top -1.27 0.49 0.2 12 0 bondPad(13) = left 0.6 0.2 0.49 pinPad(13) = top -2.54 0.49 1.0 13 0 bondPad(14) = left
0.2 0.2 1.0 pinPad(14) = top -3.81 0.49 1.0 14 0
```



Subsection

Guard Rings

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What Is Electrostatic Discharge?

Electrostatic discharge (ESD): The transfer of charge (static charges accumulated due to friction) between bodies at different electric potentials.

- The amount of charge created by contact charging is affected by the area of contact, the speed of separation, relative humidity, and so forth.
- Damage from ESD can cause:
 - Complete device failure by parametric shifts.
 - Device weakness by locally heating, melting, or otherwise damaging oxides, junctions, or device components.
- Remedies include special circuitry with ESD protection diodes, guard rings, and so forth.



Electrostatic discharge is the sudden and momentary electric current that flows between two objects at different electrical potentials. The term is usually used in the electronics and other industries to describe momentary unwanted currents that may cause damage to electronic equipment.

Electrostatic charge builds up on a chip due to improper grounding and then discharges when a low-resistance path becomes available.

ESD Structures

Three common types of models are used to evaluate the stress imposed on ESD structures:

- Human body (finger) model
- Machine model
- Charged-device model



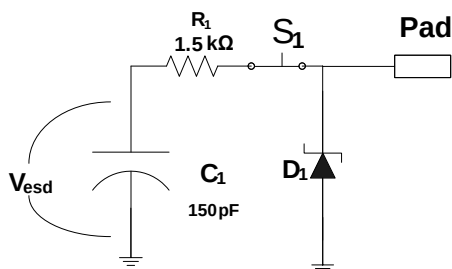
The Human Body Model is the most commonly used model for characterizing the susceptibility of an electronic device to damage from electrostatic discharge (ESD). The model is a simulation of the discharge which might occur when a human touches an electronic device.

During testing the same capacitor(modelled as 100pF) is charged to a known voltage, V_{esd} . The capacitor then discharges into ESD structure D_1 as shown in the figure through a 1500 ohm series resistor. The resistor absorbs most of the energy, but the ESD device must still conduct relatively large peak currents without failure.

Slow response, long duration and critical for all systems.

Human Body (Finger) Model

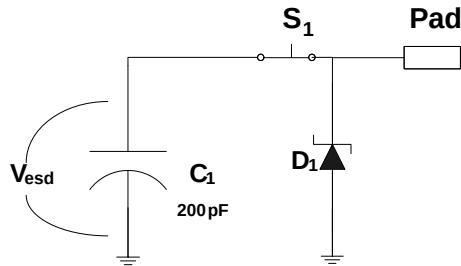
- Is modeled by a 150 pF capacitor and a 1500 Ω discharging resistance:
- Is discharged (as the fully charged capacitor during testing) through the resistor connected in series to the device under test.
- Represents a model that discharges from a human finger into a pin of a circuit package.



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Machine Model

- Is modeled by a 200 pF capacitor tied directly to the device under test.
- Simulates the discharge from a tool or machine, such as a handler or marker, to an ESD structure.
- Must not only withstand the extremely high-peak currents that result, but it also must dissipate the energy of the ESD strike without permanent damage.



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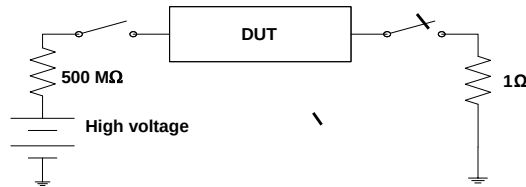
The Machine Model as shown in the figure is similar to the Human Body Model except the Machine Model contains a larger capacitor and there is no series resistor to limit the current levels. With the higher current levels in the Machine Model, most components will break down at much lower test voltages than with the Human Body Model.

The Machine Model employs a 200pF capacitor charged to a specified voltage V_{esd} . This capacitor discharges directly through ESD device D_1 . The ESD structure must not only withstand the extremely high peak currents that result, but it must dissipate the energy of the ESD strike without permanent damage.

Fast response, shorter duration and critical for only closed systems.

Charged-Device Model

- Is modeled by statically charging the component, that is, the device under test (DUT), with a high-voltage power supply through the 500 Mohm resistor to the test voltage with the connection to the 1 ohm resistor open.
- Is connected to the ground (after fully charged) through the 1 ohm resistor.
- Can be tested between any two device pins for both charge and discharge path.
- Is enabled by the 1 ohm resistor to connect to a 50 ohm scope probe to view the output waveform and measure the current levels.



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Here the DUT is modeled as a capacitor. The current levels are determined by the capacitance levels of the device under test.

This test involves charging the integrated circuit's package to a specified voltage relative to a ground plate and then discharging one pin through a low impedance probe to the plate. The charged device models generates even higher currents than the Machine model but the transient energy of the latter is lesser than the former. Because the capacitance of an integrated circuit package is much smaller than the 200pF capacitance specified by the Machine model.

DUT : device under test.

ESD Damages

- Dielectric breakdown due to high potentials.
- Gate oxides rupture at 20–25V or less.
- Polysubstrate shorts occur at 400–500V.
- Metal-metal, metal-poly breakdown occurs at less than 1000V.
- **Latent damage:** A flaw in the structure that is not apparent at the time of its onset but that reveals itself by facilitating a hard failure at a subsequent, normally nonfatal discharge stress to which the device is subjected during ordinary use.



Latent Failure

Latent failure is more difficult to identify and ultimately more costly.

Device may be partially degraded after an ESD event, remaining operable, but with compromised performance characteristics.

Devices with latent defects may experience intermittent or premature failure.

Detecting cause of failure may be difficult or hazardous.

Latent failures increase warranty and replacement costs.

ESD Protection

There are two ways to reduce IC failures caused by ESD:

- Ensure proper handling and grounding of personnel and equipment during manufacturing and usage of packaged chips to prevent ESD events from occurring.
- Connect protection circuits (almost always on-chip) to the pins of a packaged IC, which diverts high currents away from the internal circuitry and clamp high voltages during an ESD stress.

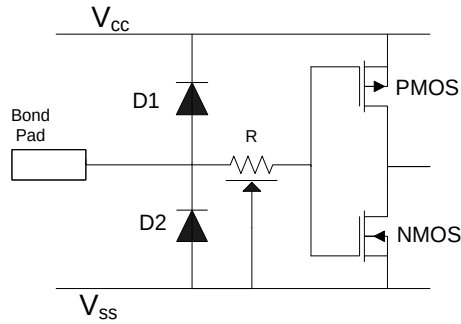


A properly implemented grounding system will prevent conductors and people from charging up to dangerous levels ($> 1\text{kV}$). It also provides (safe) discharge paths for charged objects or product.

When ESD events are suspected, several areas should be measured to verify that they are within the dissipative range.

- Check that static-dissipative clothing and wrist straps are being used and worn properly.
- Verify that static-dissipative floor and countertop mats are being used.
- Verify that machines and microscopes are properly grounded.

Diode Protection Scheme

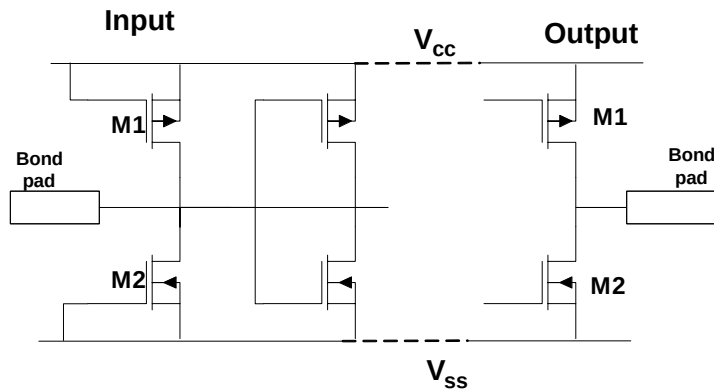


- Diode D1 becomes forward biased and conducts current for any input voltage greater than $V_{CC} + V_d$, where V_d is the forward diode drop.
- Diode D2 clamps any negative voltage below $V_{SS} - V_d$.

Parallel clamping diodes require a relatively large area and exhibit undesirable parasitic capacitance.

CMOS-Transistor Protection Scheme

- These devices can be either thin-gate or thick-gate (field) transistors.
- The drain of the NMOS device (M2) is connected to the I/O with the gate, source, and substrate tied to V_{SS} .
- A PMOS device (M1) is placed between the I/O pad and V_{CC} , with the drain connected to the I/O and the gate, source, and the substrate tied to V_{CC} .



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Source: TTM Presentation.

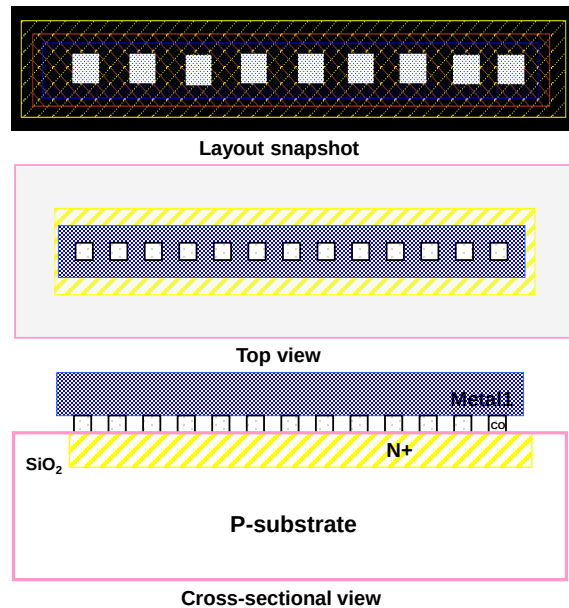
The analog macro often needs special analog I/O pads and ESD cells. The analog signals may not be able to tolerate the resistance or the capacitance of a normal I/O pin. They may also need the I/O ring separated from the ring that serves the digital sections of the chip. The I/O ring supplies can pick up a large amount of noise through capacitive coupling from the digital I/O. To maintain the ESD protection, "break cells" with anti-parallel diodes need to be inserted where the ring is cut. Special low-capacitance I/O may be required for some of the pins to maintain analog performance; and some pins (e.g., crystal oscillators are a commonly missed case) need low-leakage ESD protection. High quality macros will often include the pads and ESD structures

Guard Rings

A **guard ring** is a large, doughnut-shaped tap that prevents a phenomenon called **latchup**. When latchup occurs in an IC, it damages the device.

Guard ring diffusions should be fully strapped to the metal power supply.

The diagram shows a guard ring for a diode.



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Cadence White Papers.

Picture Source: TTM

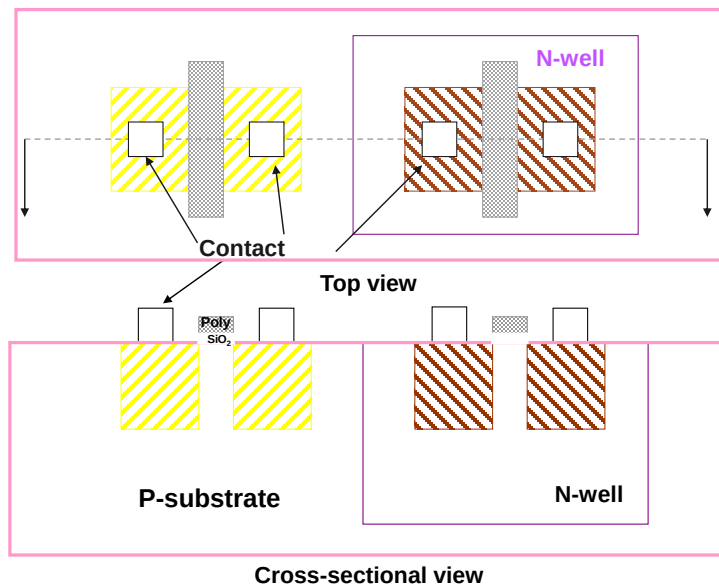
Note that additional, regular taps may still need to be placed inside a guard ring.

Omit contacts only when absolutely necessary.

Do not break the metal ring on top of the diffusion ring unless absolutely necessary.

Picture: Cadence-vda-program.pdf.

CMOS Layout



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Observe both the views.

Layout Constraints: Examples

- Layout constraints are annotations to the circuit diagram.

Layout constraints indicate how parts of the layout should be physically implemented in order to meet overall specifications of the design.

- Constraints typically cover:
 - Symmetry and matching of devices and nets
 - Maximum resistance of wires in the signal path
 - Shielding of sensitive signals
 - Guard rings
 - Power routing (minimum width and maximum resistance of bias lines)
 - Thermal issues



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Layout Constraints: Examples (continued)

- For high-speed designs, constraints may include:
 - Maximum values of parasitic capacitances, inductances, resistances
 - Inductance of bond wires (package design, but also pad placement)
 - Capacitances of the lead frame (package design)



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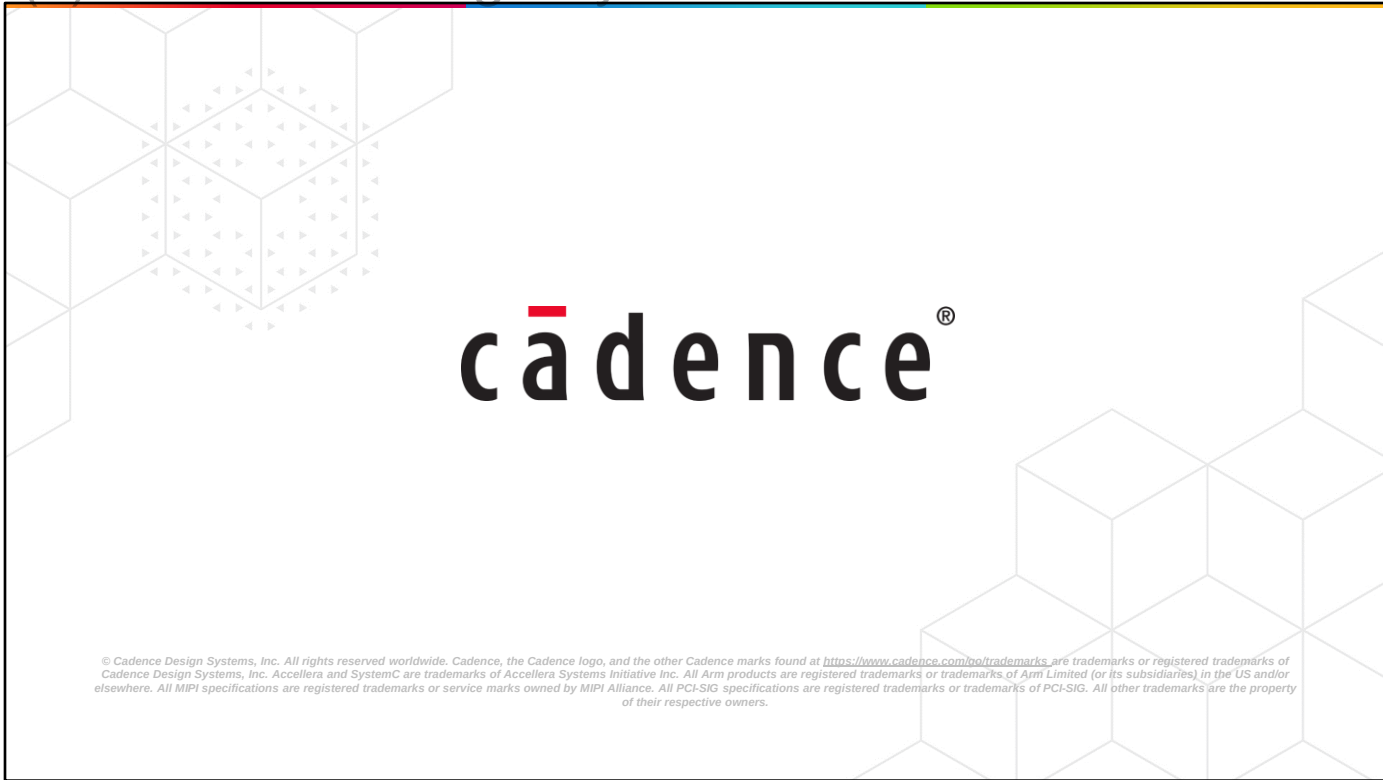
Layout Practices: Examples

Layout practices are implementation techniques not driven by constraints from the circuit designer—"This is how a good layout designer does it..."

- Grouping of related devices
- Device orientation and contact sharing
- Substrate connections and techniques for preventing latch-up and noise
- Tying I/O pins to the highest and lowest voltage on the chip in order to protect transistors from voltages outside the safe range
- Choice of metal layers for routing blocks, top level, power, ground
- Star-shaped nets to prevent offset caused by IR drop
- Via reliability
- Pinout selection



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